NEW L. J. INSTITUTE OF ENGINEERING AND TECHNOLOGY

SEMESTER: I

DEPARTMENT

OF

COMPUTER SCIENCE AND ENGINEERING

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

PRACTICE WORK BOOK

SUBJECT CODE: BE01R00081

SUBJECT NAME: BASIC MECHANICAL ENGINEERING

Subject Code:	BE01R00081	**Subject Name**	**Credit:**	4
Semester:	I	Basic Mechanical Engineering	**Hours:**	120
Department:		Computer Science and Engineering (Artificial Intelligence and Machine Learning)		

UNIT: 1

BASIC TERMINOLOGY AND ENERGY

Sub Unit 1.1 Basic Terminology

Prime movers and its types, Concept of Force, Pressure, Energy, Work, Power, System, Heat, Temperature, Specific heat capacity, Process, Cycle, Internal energy, Enthalpy, Statements of Zeroth law and First law.

TOPIC 1 – PRIME MOVER AND BASIC CONCEPTS

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Explain the following terms in brief: - (1) Heat and Work (2) Specific heat **[NLJIET]**	3	CO1	UN
2	What is Prime mover? How are they classified? **[NLJIET]**	3	CO1	UN
3	Define the following terms: (i) Prime mover (ii) Specific heat (iii) Internal energy **[NLJIET]**	3	CO1	RM
4	Discuss Closed, Open and Isolated Thermodynamic system with neat sketch. **(Jan-2019); (Nov-2020) [NLJIET]**	3	CO1	UN
5	Give the definition of (1) close system (2) open system (3) isolated system. **[NLJIET]**	3	CO1	RM
6	Define Melting point, Boiling point and Triple point of water using p-v diagram. **(Jun-2019) [NLJIET]**	3	CO1	RM
7	Define (1) Critical Point (2) Enthalpy (3) Extensive property. **(Aug-2022) [NLJIET]**	3	CO1	RM
8	What is thermal prime mover? Why they are most important prime movers? **(Aug-2022) [NLJIET]**	3	CO1	UN
9	What is stored energy and energy in transition? What is its unit? **[NLJIET]**	3	CO1	RM
10	Define Specific heat and enthalpy. **(Mar-2023) [NLJIET]**	3	CO1	RM
11	Define these terms: (1) Absolute Pressure (2) Thermal Equilibrium (3) Prime Mover. **(Jul-2024) [NLJIET]**	3	CO1	RM

12	Explain with neat sketch: Open, Close and Isolated system. **(Jan-2025-New) [NLJIET]**	**3**	**CO1**	**UN**
13	Discuss Closed, Open and Isolated Thermodynamic system with neat sketch. **(Jun-2025-New) [NLJIET]**	**3**	**CO1**	**UN**
14	Explain open system, closed system and isolated system. **(Jul-2025) [NLJIET]**	**3**	**CO1**	**UN**
15	Define Pressure and explain Absolute Pressure, Gauge Pressure and Atmospheric pressure. **[NLJIET]**	**4**	**CO1**	**RM**
16	Define following terms (i) Absolute pressure and Atmospheric pressure (ii) Enthalpy and Energy. **[NLJIET]**	**4**	**CO1**	**RM**
17	Differentiate between: 1. Open and Close system. 2. Intensive and Extensive properties. **[NLJIET]**	**4**	**CO1**	**UN**
18	Explain with suitable examples the following thermodynamic systems: (1) Isolated (2) Open. **(Jul-2024) [NLJIET]**	**4**	**CO1**	**UN**
19	Name the types of prime movers based on the source of energy. **(Jul-2025) [NLJIET]**	**4**	**CO1**	**UN**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Define the following terms: Prime mover, Boundary, Latent Heat, Temperature, First law of thermodynamics. **[NLJIET]**	**5**	**CO1**	**RM**
2	How prime movers are classified? Explain different sources of energy used by them. **[NLJIET]**	**7**	**CO1**	**UN**
3	Define prime mover and classify it. **(Jan-2025) [NLJIET]**	**7**	**CO1**	**UN**
4	Give S.I. Units of Followings: (1) Work (2) Enthalpy (3) Heat (4) Power (5) Force (6) Energy (7) Specific heat (8) Specific volume. **[NLJIET]**	**8**	**CO1**	**RM**

NUMERICALS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Determine the amount of heat required to raise the temperature of a steel work piece from 40 to 160 deg C. The mass of work piece is 20 kg and specific heat is 460 J/kg K. **(Jul-2023) [NLJIET]**	**3**	**CO1**	**AN**

2	A thermodynamic cycle has four processes. The heat and work interaction with surroundings is given in the table. Determine the work done during process 3-4. What will be the change in internal energy after completion of one cycle? **(Jul-2023) [NLJIET]**	4	CO1	AN
3	Determine the power required to accelerate a 900 kg car from rest to a velocity of 80 km/h in 20 sec on a horizontal road. (Neglect friction, rolling resistance, air drag, etc.) **(Jan-2024) [NLJIET]**	4	CO1	AN

Process	Heat Transfer (kJ)	Work done (kJ)
1-2	+925	+70
2-3	-110	-50
3-4	-770	?
4-1	+220	+170

TOPIC 2 – LAWS OF THERMODYNAMICS

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Define Zeroth law of thermodynamics and First law of thermodynamics. **(Mar-2022) [NLJIET]**	3	CO1	RM
2	Give comparison between work and heat. **(Aug-2022) [NLJIET]**	3	CO1	UN
3	State and explain zeroth law of thermodynamics. **(Jan-2025) [NLJIET]**	3	CO1	RM
4	State and explain Zeroth law of thermodynamics and First law of thermodynamics. **(Jun-2025-New) [NLJIET]**	3	CO1	RM
5	Differentiate between heat and work in the context of the First Law. **(Jul-2025) [NLJIET]**	3	CO1	UN
6	State and explain First law of thermodynamics. **[NLJIET]**	4	CO1	RM
7	Write only the statement of (i) Zeroth law (ii) First and Second law of Thermodynamics. **[NLJIET]**	4	CO1	RM
8	Write similarities between heat transfer & work transfer. **[NLJIET]**	4	CO1	UN
9	Explain Specific heat. Give Statements of Zeroth Law and First law of thermodynamics. **[NLJIET]**	4	CO1	UN

10	Define the heat and give the comparison of heat and work. **[NLJIET]**	**4**	**CO1**	**RM**
11	Write the mathematical form of the First Law and explain each term. **(Jul-2025) [NLJIET]**	**4**	**CO1**	**RM**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Define the following: - 1. Energy 2. Specific heat 3. Zeroth law of thermodynamics 4. Temperature. **[NLJIET]**	**7**	**CO1**	**RM**
2	Define Zeroth law, First law and Second law of thermodynamics. Give limitations of First law of thermodynamics. **[NLJIET]**	**7**	**CO1**	**RM**
3	Define the following terms: Prime movers, Internal energy, Heat capacity, First law of thermodynamics, State, Path, Zeroth law of thermodynamics. **(Mar-2021) [NLJIET]**	**7**	**CO1**	**RM**

UNIT: 1

BASIC TERMINOLOGY AND ENERGY

Sub Unit 1.2 Energy

Applications of Energy sources like Fossil fuels, Nuclear fuels, Hydrogen fuel, Hydro, Solar, Wind, and Bio-fuels, Environmental issues like Global warming and Ozone depletion.

TOPIC 1 – INTRODUCTION TO FUELS

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Define Calorific value and explain Higher and Lower Calorific values. **[NLJIET]**	**3**	**CO1**	**RM**
2	Give advantages of liquid fuels compared to solid fuels. **(Aug-2022) [NLJIET]**	**4**	**CO1**	**RM**
3	What are LPG and CNG? **[NLJIET]**	**4**	**CO1**	**RM**
4	What is solid fuel? Discuss different types of solid fuel. **(Jan-2019) [NLJIET]**	**4**	**CO1**	**RM**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	State the advantages of gaseous fuels over solid and liquid fuels. **[NLJIET]**	5	CO1	RM
2	List various gaseous fuels. State its advantages and disadvantages. **[NLJIET]**	6	CO1	RM
3	List the various liquid fuels. State its advantages over solid fuels. **[NLJIET]**	6	CO1	RM
4	Explain about different types of fuels. **[NLJIET]**	7	CO1	UN
5	State the advantages and disadvantages of gaseous fuels. List the various gaseous fuels. **[NLJIET]**	7	CO1	RM
6	Write advantages of gaseous fuels over other fuels. Write short note on CNG. **[NLJIET]**	7	CO1	RM
7	Give detail classification of fuel and write short note on solid fuel. **[NLJIET]**	7	CO1	UN
8	Which are common solid fuels? Write in brief about each of them. Define Calorific value of Fuel. **[NLJIET]**	7	CO1	UN

TOPIC 2 – NATURAL ENERGY AND POLLUTION

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	What do you mean by non-conventional energy sources? How does it differ from conventional sources? **[NLJIET]**	3	CO1	UN
2	Write a short note on Bio-fuels. **[NLJIET]**	3	CO1	UN
3	What are the various forms of energy? List the non-convectional sources of energy. **[NLJIET]**	3	CO1	UN
4	Explain any one Renewable source of energy. **[NLJIET]**	3	CO1	UN
5	Discuss the factors responsible for global warming and ozone depletion. **(Jun-2019) [NLJIET]**	3	CO1	UN
6	List advantages of Solar energy and its application. **(Mar-2021) [NLJIET]**	3	CO1	RM
7	Write a short note on “Global Warming and solar energy”. **(Jul-2025) [NLJIET]**	3	CO1	UN

8	Write a short note on "Global Warming and Ozone depletion". **[NLJIET]**	**4**	**CO1**	**UN**
9	Describe any four forms of energy in 100 words. **[NLJIET]**	**4**	**CO1**	**UN**
10	Explain advantages and disadvantages of wind energy. **(Jan-2024) [NLJIET]**	**4**	**CO1**	**UN**
11	Explain phenomena of global warming and causes of ozone depletion. **(Mar-2021) [NLJIET]**	**4**	**CO1**	**UN**
12	Write a short note on "Global Warming" and "Solar energy". **(Mar-2022) [NLJIET]**	**4**	**CO1**	**UN**
13	Explain difference between renewable and non-renewable energy. **(Mar-2023) [NLJIET]**	**4**	**CO1**	**UN**
14	Explain difference between renewable and non-renewable energy. **(Jun-2025-New) [NLJIET]**	**4**	**CO1**	**UN**
15	Make a list of 1. Natural fuels and 2. Artificial (prepared) fuels. Write two effective points to reduce global warming. **(Jul-2023) [NLJIET]**	**4**	**CO1**	**UN**
16	List sources of renewable energy. **(Jan-2025) [NLJIET]**	**4**	**CO1**	**UN**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	What are Bio-fuels? Describe them in details. **[NLJIET]**	**5**	**CO1**	**UN**
2	Give detailed classification of fuel. Write short note on wind energy. **[NLJIET]**	**7**	**CO1**	**UN**
3	List the non-convectional sources of energy. Prepare a short note on solar energy. **(Jan-2020) [NLJIET]**	**7**	**CO1**	**UN**
4	Write a short note on "Renewable Energy Sources". **(Jul-2024) [NLJIET]**	**7**	**CO1**	**UN**
5	Discuss the primary causes of global warming. What are impacts of global warming? **(Jan-2025-New) [NLJIET]**	**7**	**CO1**	**UN**

UNIT: 2

PROPERTIES OF GASES AND PROPERTIES OF STEAM

Sub Unit 2.1 Properties of Gases

Boyle's law, Charles's law, Gay-Lussac's law, Avogadro's law, Combined gas law, Gas constant, Relation between Cp and Cv, Various non-flow processes like Constant volume process, Constant pressure process, Isothermal process, Adiabatic process, Polytrophic process.

TOPIC 1 – DIFFERENT GAS LAWS, CHARACTERISTIC GAS EQUATION, RELATION BETWEEN Cp AND Cv

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Write about Combined gas law, Gas constant and Non-flow process. **[NLJIET]**	3	CO2	UN
2	Prove that the difference between two specific heats of gases is equal to its characteristic gas constant. **(Jan-2020) [NLJIET]**	3	CO2	AP
3	State the following: Charle's Law, Boyle's Law, Characteristic gas equation. **[NLJIET]**	3	CO2	UN
4	Define the following terms: Boyle's law, Avogadro's law, Charle's law. **(Mar-2021) [NLJIET]**	3	CO2	UN
5	Derive characteristic equation of a perfect gas. **(Jan-2025) [NLJIET]**	3	CO2	AP
6	Write the statements for Boyle's law, Charles's law and Gay-Lussac's law. **(Jan-2025-New) [NLJIET]**	3	CO2	RM
7	With usual notations prove that Cp – Cv = R. **(Jul-2025) [NLJIET]**	3	CO2	AP
8	Differentiate between Gas constant and Universal gas constant. **(Jan-2020) [NLJIET]**	4	CO2	UN
9	With usual notations prove that Cp – Cv= R. **(Jun-2025-New) [NLJIET]**	4	CO2	AP

DESCRIPTIVE QUESTIONS				
Sr. No.	**QUESTIONS**	**Marks**	**CO**	**RBT Level**
1	Derive expression PV / T = constant with the help of Boyle's law and Charle's law. **[NLJIET]**	**5**	**CO2**	**AP**
2	Define specific heat at Constant volume, Constant pressure and Adiabatic index. Also derive relationship between specific heats in form of Characteristic gas constant. **(Jun-2019) [NLJIET]**	**7**	**CO2**	**AP**
3	Define perfect gas and with usual notation prove Cp - Cv = R for perfect gas. **[NLJIET]**	**7**	**CO2**	**AP**
4	Derive the characteristics gas equation for a perfect gas with usual notations. **(Aug-2022) [NLJIET]**	**7**	**CO2**	**AP**
5	Derive the characteristics gas equation for a perfect gas with usual notations. **(Jun-2025-New) [NLJIET]**	**7**	**CO2**	**AP**
6	Derive the characteristics gas equation for a perfect gas with usual notations. **(Jul-2025) [NLJIET]**	**7**	**CO2**	**AP**

NUMERICALS				
Sr. No.	**QUESTIONS**	**Marks**	**CO**	**RBT Level**
1	A vessel of 2.5 m^3 capacity contains one kg mole of nitrogen at 100 deg C. Calculate the specific volume and pressure if the gas is cooled to 30 deg C. Calculate final pressure, change in specific internal energy and specific enthalpy. **(Mar-2021) [NLJIET]**	**4**	**CO2**	**AN**
2	An ideal gas is heated from 25 deg C to 145 deg C. The mass of gas is 2 kg. Determine (i) Specific heats (ii) Change in internal energy, (iii) Change in enthalpy. Assume R = 267 J/kg.K and γ =1.4 for the gas. **[NLJIET]**	**7**	**CO2**	**AN**
3	A gas initially at 510 kPa and volume of 142 liters undergoes a certain process after which the pressure and volume reach to 170 kPa and 275 liters respectively. And there is enthalpy drop of 65 kJ. The specific heat at constant volume for this gas is 718 J/kg-K. Calculate: 1. Change in internal energy; 2. Specific heat at constant pressure; 3. Gas Constant. **(Jul-2024) [NLJIET]**	**7**	**CO2**	**AN**

4	5 kg of air is heated from 25 deg C to 150 deg C. Determine: (i) specific heats, (ii) change in internal energy, (iii) change in enthalpy and (iv) heat supplied. Assume R = 0.287 kJ/kg.K and γ = 1.4 for air and work done = 500 kJ. **(Jan-2025-New) [NLJIET]**	7	CO2	AN
5	One kg of an ideal gas is heated from 18 deg C to 98 deg C assuming R = 0.27 kJ/kg.K and γ = 1.18 for gas, calculate (1) specific heats (Cp and Cv) (2) Change in internal energy (3) change in enthalpy. **(Jun-2025-New) [NLJIET]**	7	CO2	AN

TOPIC 2 – NON FLOW PROCESSES :- CONSTANT VOLUME PROCESS AND CONSTANT PRESSURE PROCESS

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	What is flow and non-flow process? **[NLJIET]**	3	CO2	RM
2	Represent following process with specification on Pressure-Volume diagram for same conditions (1) Constant Pressure process (2) Constant Volume process (3) Isothermal Process. **[NLJIET]**	3	CO2	RM

NUMERICALS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	A gas whose pressure, volume, and temperatures are 2.75 bar, 0.09 m^3 and 185 deg C respectively has the state changed at constant pressure until its temperature becomes 15 deg C. Calculate (i) Heat Transferred. (ii) Work Done during the process. Take R = 0.29 kJ/kg.K and Cp = 1.005 kJ/kg.K **(Jan-2024) [NLJIET]**	7	CO2	AN
2	A cylindrical vessel of 1 m diameter and 4 m length has hydrogen gas at pressure of 100 kPa and 27 deg C. Determine the amount of heat to be supplied so as to increase pressure to 125 kPa. For hydrogen, take Cp = 14.307 kJ/kg.K, Cv = 10.183 kJ/kg.K **(Aug-2022) [NLJIET]**	7	CO2	AN

3	A gas whose pressure, volume and temp. are 3 bar, 0.1 m^3 and 190 deg C respectively has the state changed at constant pressure until the temperature becomes 15 deg C. Calculate (i) Heat transferred. (ii) Work done during the process. Take R = 0.29 kJ/kg.K & Cp = 1.005 kJ/kg.K **[NLJIET]**	**7**	**CO2**	**AN**
4	Air has a volume of 0.15 m^3, pressure 1.5 bar and temperature 107 deg C. It is compressed at constant pressure, until its volume becomes 0.11 m^3. Determine: (i) Temperature at the end of compression, (ii) Work done during compression, (iii) Change in internal energy. Take Cp = 1.005 kJ/kg.K, Cv = 0.718 kJ/kg.K **(Jul-2023) [NLJIET]**	**7**	**CO2**	**AN**
5	The gauge pressure of an automobile tire is measured to be 210 kPa before a trip and 220 kPa after the trip at a location where the atmospheric pressure is 95 kPa. Assuming the volume of the tire remains constant and the air temperature before the trip is 25 deg C, determine air temperature in the tire after the trip. Convert this change in temperature into percentage change. **(Jan-2024) [NLJIET]**	**7**	**CO2**	**AN**

TOPIC 3 – NON FLOW PROCESS :- CONSTANT TEMPERATURE PROCESS

SHORT QUESTION

Sr. No.	QUESTION	Marks	CO	RBT Level
1	Define isothermal process. Derive the expression for work done and heat transfer for this process. **[NLJIET]**	**4**	**CO2**	**AP**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	What is isothermal process? Derive an expression for the work done during the isothermal process. **[NLJIET]**	**5**	**CO2**	**AP**
2	Describe Isothermal process and derive expression for Work done, Change in Internal energy, Heat transfer and Change in Enthalpy. **(Jan-2019) [NLJIET]**	**7**	**CO2**	**AP**

NUMERICALS				
Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	One kg of gas at a pressure of 100 kPa and temperature of 17 deg C is compressed isothermally in a piston-cylinder arrangement to final pressure of 2500 kPa. The characteristic gas equation is given by the relation pV = 260T per kg where T is in Kelvin. Find out (1) Final Volume (2) Compression Ratio (3) Change in enthalpy (4) Work done on the gas. **(Jan-2020) [NLJIET]**	7	CO2	AN
2	In air compressor, air enters at 1.013 bar and 27 deg C having volume of 5 m^3/ kg and it is compressed to 12 bar isothermally. Determine (i) work done (ii) heat transfer, and (iii) change in internal energy. **[NLJIET]**	7	CO2	AN
3	One kg of gas at 1 bar pressure and 17 deg C is compressed isothermally to a pressure of 25 bar in a cylinder. The characteristic gas constant is 260 J/kg.K Calculate (a) The final temperature (b) Work done and (c) Change in enthalpy. **(Mar-2022) [NLJIET]**	7	CO2	AN

TOPIC 4 – NON FLOW PROCESS :- ADIABATIC PROCESS

SHORT QUESTION				
Sr. No.	QUESTION	Marks	CO	RBT Level
1	Find expression of work done and change in internal energy for adiabatic process. **(Jan-2025-New) [NLJIET]**	4	CO2	AP

DESCRIPTIVE QUESTIONS				
Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Define adiabatic process. Prove that PV^γ = Constant. **(Mar-2023) [NLJIET]**	7	CO2	AP
2	Define adiabatic process. Derive the relation between P, V and T for this process. Also derive the expression for work done and change in internal energy for this process. **[NLJIET]**	7	CO2	AP

3	For adiabatic process, derive relation between pressure, volume and temperature. Also derive equation for work done and heat transfer. **[NLJIET]**	**7**	**CO2**	**AP**
4	Explain adiabatic process. Derive an expression for work done during the adiabatic expansion of an ideal gas. **(Mar-2021) [NLJIET]**	**7**	**CO2**	**AP**

NUMERICALS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	One kg gas is compressed adiabatically by following the law $pV^{\gamma} = C$ from initial temperature of 290 K. The initial pressure of gas is 1 bar. The initial and final volumes of gas are 0.50 m^3 and 0.05 m^3 respectively. Find the final temperature and pressure of gas. Assume γ =1.4 **(Jan-2020) [NLJIET]**	**3**	**CO2**	**AN**
2	Determine the work done in compressing one kg of air from a volume of 0.15 m^3 at a pressure of 1.0 bar to a volume of 0.05 m^3 when the compression is (1) Isothermal and (2) Adiabatic, Take γ =1.4. Also, comment on your answer. **[NLJIET]**	**7**	**CO2**	**AN**
3	The initial volume of 0.9 kg of a certain gas was 0.75 m^3 at a temperature of 15 deg C and a pressure of 1 bar. After adiabatic compression, the volume is reduce to 0.28 m^3 and pressure was found to be 4 bar. Calculate (1) Gas constant (2) Molecular mass if R_0 = 8314.3 J/kg.mol.K, (3) Ratio of specific heats (4) Cp and Cv (5) Change in internal energy. **(Aug-2022) [NLJIET]**	**7**	**CO2**	**AN**
4	0.35 m^3 of gas at 10 bar, 157 deg C expands adiabatically to 4 bar. It is then compressed isothermally to its original volume. Find the final temperature and pressure of the gas. Take Cp = 0.996 kJ/kg.K; Cv = 0.703 kJ/kg.K **(Jul-2023) [NLJIET]**	**7**	**CO2**	**AN**

TOPIC 5 – NON FLOW PROCESS :- POLYTROPIC PROCESS NUMERICALS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	One kg of gas is compressed polytropically from 150 kPa pressure and 290 K temperature to 750 kPa. The compression is according to law $PV^{1.3}$ = constant. Find: (a) Final temperature (b) Work-done (c) Change in internal energy (d) Amount of heat transfer and (e) Change in enthalpy. Take R = 0.287 kJ/kg.K and Cp = 1.001 kJ/kg.K **[NLJIET]**	7	CO2	AN
2	0.67 kg of gas at 14 bar and 290 deg C is expanded to four times the original volume according to the law $PV^{1.3}$ = Constant. Calculate: (1) The original and final volume of the gas. (2) The final temperature of the gas. (3) The final pressure of the gas. Take R = 287 J/kg.K **[NLJIET]**	7	CO2	AN
3	1 kg of air at 9 bar pressure and 80 deg C temperature undergoes a non-flow work polytropic process. The law of expansion is $PV^{1.1}$ = C. The pressure falls to 1.4 bar during process. Calculate (1) Final temperature (2) Work done (3) Change in internal energy (4) Heat exchange. Take R = 287 J/kg.K and γ = 1.4 for air. **[NLJIET]**	7	CO2	AN
4	One kg of gas is compressed polytropically from 160 kPa pressure and 280 K temperature to 760 kPa. The compression is according to law $PV^{1.3}$= Constant. Find: (1) Final Temperature (2) Work done (3) Change in internal energy (4) Amount of heat transfer and (5) Change in enthalpy. Take R = 0.287 kJ/kg.K and Cp = 1.002 kJ/kg.K **[NLJIET]**	7	CO2	AN
5	One cubic meter of air at pressure of 1.5 bar and 80 deg C is compressed to final pressure 8 bar and volume 0.28 m^3. Determine: - 1. Mass of air 2. Index of compression 'n' 3. Change in internal energy 4. Heat transfer during compression. Take γ = 1.4 and R= 287 J/kg.K **[NLJIET]**	7	CO2	AN

6	A cylinder contains 0.6 m^3 of a gas at a pressure of 1 bar and 90 deg C. The gas is compressed to a volume of 0.18 m^3 by the law $PV^n = C$. The pressure of gas at the end of compression is 5 bar. Calculate: 1. Mass of gas 2. Value of index ‘n’ 3. The change in internal energy of the gas. 4. Work done 5. The heat received or rejected by the gas during the process. Take γ =1.4 and R= 0.294 kJ/kg.K **[NLJIET]**	**7**	**CO2**	**AN**
7	A cylinder contains 0.5 m^3 of a gas at a pressure of 1 bar and 90 deg C. The gas is compressed to a volume of 0.15 m^3. The final pressure is 5 bar. Calculate: (i) The mass of gas (ii) Value of index ‘n’ for compression (iii) The increase in internal energy. Take γ = 1.4 and R = 294.2 J/kg.K **[NLJIET]**	**7**	**CO2**	**AN**
8	2 kg of gas is compressed from 1 bar and 30 deg C according to the law $PV^{1.3}$ = C. The pressure after compression is 6 bar. Calculate (1) Characteristic Gas Constant (2) Temperature at the end of compression (3) Work done (4) Heat supplied (5) Change in Internal Energy. Take molecular mass of gas = 30, Cp = 1.75 kJ/kg.K **[NLJIET]**	**7**	**CO2**	**AN**
9	1 kg of air at 7 bar pressure and 90 deg C temperature undergoes a non-flow polytropic process. The law of expansion is $PV^{1.1}$ = C. The pressure falls to 1.4 bar during process. Calculate (1) Final temperature (2) Work done (3) Change in internal energy (4) Heat exchange. Take R = 287 J/kg.K and γ = 1.4 for air. **(Mar-2023) [NLJIET]**	**7**	**CO2**	**AN**
10	One kg of air at 7 bar pressure and 90 deg C temperature undergoes a non-flow process. The law of expansion is $PV^{1.1}$ = C. The pressure falls to 1.4 bar during process calculate (1) the final temperature (2) work done (3) Change in internal energy (4) Heat exchange. Take R= 287 J/kg.K and γ=1.4 for air. **(Jun-2025-New) [NLJIET]**	**7**	**CO2**	**AN**

11	One kg of gas is compressed polytropically from 150 kPa pressure and 290 K temperature to 750 kPa. The compression is according to law $pV^{1.3} = C$. find: Final temperature; Work-done; Change in internal energy; Amount of heat transferred; Change in enthalpy. Take R=0.287 kJ/kg K and Cp = 1.001 kJ/kg.K **(Jan-2025) [NLJIET]**	**7**	**CO2**	**AN**

UNIT: 2

PROPERTIES OF GASES AND PROPERTIES OF STEAM

Sub Unit 2.2 Properties of Steam

Steam formation, Types of steam, Enthalpy, Specific volume, Internal energy and dryness fraction of steam, use of steam tables, steam calorimeters.

TOPIC 1 – BASIC CONCEPTS OF STEAM

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Define following terms with respect to steam (i) Dryness fraction (ii) Degree of super heat (iii) Specific volume. **[NLJIET]**	**3**	**CO2**	**RM**
2	Define the following terms: (a) Wet steam (b) Degree of superheat (c) Saturation temperature. **[NLJIET]**	**3**	**CO2**	**RM**
3	Express the mathematical formula with standard notation/symbol of properties for 1. Wetness fraction of steam 2. Enthalpy of superheated steam 3. Specific volume of wet steam **[NLJIET]**	**3**	**CO2**	**AP**
4	Describe the process of steam formation on T-H diagram. **(Mar-2022) [NLJIET]**	**3**	**CO2**	**RM**
5	Dryness fraction of steam cannot have the value more than unity. Justify. **(Jan-2019) [NLJIET]**	**3**	**CO2**	**AP**
6	What is superheated steam; list its advantages and application. **(Mar-2021) [NLJIET]**	**3**	**CO2**	**RM**
7	Explain - Dryness fraction of steam. **(Mar-2023) [NLJIET]**	**3**	**CO2**	**UN**
8	What is dryness fraction? Also state its importance. **(Jun-2025-New) [NLJIET]**	**3**	**CO2**	**UN**
9	Prove that dryness fraction + wetness fraction = 1. **[NLJIET]**	**4**	**CO2**	**AP**

10	Define: (i) Sensible heat (ii) Enthalpy of evaporation (iii) Heat of superheat (iv) Dryness Fraction. **[NLJIET]**	**4**	**CO2**	**RM**
11	Define: (i) Sensible heat (ii) Latent heat (iii) Dryness fraction (iv) Enthalpy of evaporation. **[NLJIET]**	**4**	**CO2**	**RM**
12	Define- (i) Dryness fraction, (ii) Wetness fraction. **(Mar-2022) [NLJIET]**	**4**	**CO2**	**RM**
13	Define (1) Sensible heat (2) Latent heat (3) Degree of superheat (4) Specific volume of steam. **(Aug-2022) [NLJIET]**	**4**	**CO2**	**RM**
14	Write the uses of "Steam Tables". **(Nov-2020) [NLJIET]**	**4**	**CO2**	**UN**
15	Define following terms: 1. Dryness fraction 2. Degree of superheat 3. Enthalpy of evaporation 4. Enthalpy of superheated steam. **(Jan-2025) [NLJIET]**	**4**	**CO2**	**RM**

DESCRIPTIVE QUESTION

Sr. No.	QUESTION	Marks	CO	RBT Level
1	Explain steam formation with T-H diagram. **[NLJIET]**	**7**	**CO2**	**UN**

NUMERICALS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	A sample of wet steam at a pressure of 25 bar absolute has dryness fraction 0.80. Determine its enthalpy and internal energy. **[NLJIET]**	**4**	**CO2**	**AN**
2	Determine the condition of steam at a pressure of 10 bar and temperature of 200 deg C. **(Jul-2024) [NLJIET]**	**4**	**CO2**	**AN**
3	Determine the dryness fraction of steam after throttling from pressure 14 bar to 1.1 bar. The initial dryness fraction of steam is equal to 0.70. **(Jul-2024) [NLJIET]**	**4**	**CO2**	**AN**
4	Determine the mass of 0.15 m^3 of wet steam at a pressure of 4 bar and dryness fraction 0.8. Also calculate the heat of 1 m^3 of steam. **[NLJIET]**	**7**	**CO2**	**AN**

5	Determine the enthalpy and internal energy of 1 kg of steam at a pressure 10 bar (abs.), (i) When the dryness fraction of the steam is 0.85. (ii) When the steam is dry and saturated. (iii) When the steam is superheated to 300 deg C. Neglect the volume of water and take the specific heat of superheated steam as 2.1 kJ/kg.K. **[NLJIET]**	**7**	**CO2**	**AN**
6	What is a superheated steam? How much heat is added to convert 3 kg of water at 30 deg C into steam at 8 bar and 210 deg C? Take specific heat of superheated steam as 2.1 kJ/kg-K and that of water as 4.186 kJ/kg-K **[NLJIET]**	**7**	**CO2**	**AN**
7	Dry saturated steam at 7 bar pressure is expanded to 1 bar following the law $PV^{1.1}$=constant. Determine (i) Work-done (ii) Change in internal energy (iii) Heat transferred during the process. **[NLJIET]**	**7**	**CO2**	**AN**
8	Calculate the enthalpy per kg of steam at 10 bar pressure and a temperature of 300 deg C. Find also the change in enthalpy if this steam is expanded to 1.4 bar and dryness fraction of 0.8. Take specific heat of superheat steam equal to 2.29 kJ/kg.K **[NLJIET]**	**7**	**CO2**	**AN**
9	How much heat to be added to convert 4 kg of water at 20 deg C into steam at 8 bar and 200 deg C. Take Cp of superheated steam as 2.1 kJ/kg.K and specific heat of water as 4.187 kJ/kg.K **(Mar-2023) [NLJIET]**	**7**	**CO2**	**AN**
10	How much heat to be added to convert 4 kg water at 20 deg C into steam at 8 bar and 200 deg C. Take Cp of superheated steam as 2.1 kJ/kg.K and specific heat of water of water as 4.187 kJ/kg.K. **(Jul-2025) [NLJIET]**	**7**	**CO2**	**AN**
11	Find the internal energy of 1 kg of steam at a pressure of 15 bar when steam is (1) Super-heated at a temperature of 400 deg C and (2) Wet with dryness fraction of 0.9. Take specific heat of superheated steam as 2.1 kJ/kg-K. **(Jan-2020) [NLJIET]**	**7**	**CO2**	**AN**

12	1.5 kg of steam at a pressure of 10 bar and temperature of 250 deg C is expanded until the pressure becomes 2.8 bar. The dryness fraction of steam is then 0.9. Calculate change in internal energy. **(Nov-2020) [NLJIET]**	**7**	**CO2**	**AN**
13	What amount of heat is required to produce 5 kg of steam at a pressure of 5 bar and temperature of 250 deg C from water at 30 deg C. Take Cps = 2.1 kJ/kg.K **[NLJIET]**	**7**	**CO2**	**AN**
14	Determine the quality of steam for the following cases: (i) P=10 bar, v=0.180 m^3/kg (ii) P=10 bar, T=200 deg C (iii) P=25 bar, h=2750 kJ/kg **[NLJIET]**	**7**	**CO2**	**AN**
15	Calculate the total amount of heat required to produce 5 kg of steam at a pressure of 5 bar and temperature of 250 deg C from the water at 25 deg C. Take specific heat of steam = 2.1 kJ/kg-K and the specific heat of water = 4.187 kJ/kg-K **[NLJIET]**	**7**	**CO2**	**AN**
16	3 kg of steam at pressure of 10 bar exists in the following conditions. Calculate its enthalpy in each of the following cases: (1) Steam with dryness fraction = 0.91. (2) Steam at temperature 200 deg C (3) Dry and Saturated steam. **[NLJIET]**	**7**	**CO2**	**AN**
17	Determine the value of final dryness fraction of steam. 1. After losing 125 kJ from the steam at constant pressure. 2. After expansion to 3 bar pressure in a turbine stage and work equivalent of 20 kJ/kg done. Initially steam is available at 7 bar pressure and 0.9 dryness fraction. **(Jun-2019) [NLJIET]**	**7**	**CO2**	**AN**
18	Three kg of steam at a pressure of 10 bar exists in the following conditions. Calculate its enthalpy and internal energy in each of the cases. (1) Steam with x = 0.91. (2) Steam at temperature 200 deg C. **(Jun-2019) [NLJIET]**	**7**	**CO2**	**AN**
19	Calculate the internal energy of 1 kg of superheated steam at a pressure of 10 bar and 280 deg C. If this steam is to be expanded to a pressure of 1.6 bar and 0.8 kg dry, determine the change in internal energy. Assume specific heat of superheated steam as 2.1 kJ/kg.K **(Mar-2021) [NLJIET]**	**7**	**CO2**	**AN**

<table>
<tr><td>20</td><td>Steam is initially dry saturated at 9 bar in each of the following cases. Determine,
(1) Its dryness fraction if it loses 50 kJ/kg heat at constant pressure,
(2) The degree of superheat if it receives 150 kJ/kg heat at constant pressure. (Jul-2023) [NLJIET]</td><td>7</td><td>CO2</td><td>AN</td></tr>
<tr><td>21</td><td>Determine the missing properties and the phase descriptions in the following table for water. Use steam tables and write necessary explanations. (Jan-2024) [NLJIET]
<table>
<tr><td></td><td>T, deg C</td><td>P, kPa</td><td>h, kJ/kg</td><td>x</td><td>Phase description</td></tr>
<tr><td>(i)</td><td></td><td>200</td><td></td><td>0.6</td><td></td></tr>
<tr><td>(ii)</td><td>124</td><td></td><td>1835.2</td><td></td><td></td></tr>
</table></td><td>7</td><td>CO2</td><td>AN</td></tr>
<tr><td>22</td><td>A sample of wet steam at pressure of 15 bar has a dryness fraction of 0.06. Determine the enthalpy, volume, entropy and internal enthalpy per kg. (Jun-2025-New) [NLJIET]</td><td>7</td><td>CO2</td><td>AN</td></tr>
</table>

TOPIC 2 – STEAM CALORIMETERS

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	What is Throttling calorimeter? Explain its limitation. **(Nov-2020) [NLJIET]**	**3**	**CO2**	**UN**
2	What is throttling calorimeter? Also state its limitation. **(Jun-2025-New) [NLJIET]**	**3**	**CO2**	**UN**
3	Explain with neat sketch, Throttling calorimeter. **(Mar-2023) [NLJIET]**	**3**	**CO2**	**UN**
4	Draw neat diagram of the Separating Calorimeter. **(Jul-2024) [NLJIET]**	**3**	**CO2**	**RM**
5	With neat sketch, explain construction and working of Separating calorimeter. **(Mar-2021) [NLJIET]**	**4**	**CO2**	**UN**
6	Explain Barrel calorimeter with neat sketch. **[NLJIET]**	**4**	**CO2**	**UN**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	With a neat sketch, explain structure and working of Combined separating and throttling calorimeter and derive the suitable equation for finding the dryness fraction of steam. **[NLJIET]**	7	CO2	UN
2	What is throttling process? Explain Throttling calorimeter with neat sketch. Derive equation for dryness fraction. **[NLJIET]**	7	CO2	UN
3	Explain with neat sketch Throttling calorimeter. Also state its advantages and disadvantages. **(Aug-2022) [NLJIET]**	7	CO2	UN
4	Write a short note on Separating calorimeter with its limitations. **[NLJIET]**	7	CO2	UN
5	Draw a neat, labeled diagram of Barrel calorimeter. Write the equation used for this calorimeter for finding dryness fraction and state what each term in it indicates. **(Jan-2024) [NLJIET]**	7	CO2	UN
6	List methods of measuring dryness fraction. Explain any one of them. **[NLJIET]**	7	CO2	UN
7	With a neat sketch explain Throttling calorimeter. **(Jan-2025-New) [NLJIET]**	7	CO2	UN
8	Explain with neat sketch throttling calorimeter. Also state its advantages and disadvantages. **(Jul-2025) [NLJIET]**	7	CO2	UN

NUMERICALS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Combined separating and throttling calorimeter is used to find out dryness fraction of steam. Following readings were taken: Main pressure = 12 bar ab. Mass of water collected in separating calorimeter = 2 kg. Mass of steam condensed in separating calorimeter = 20 kg. Temperature of steam after throttling = 110 deg C. Pressure of steam after throttling = 1 bar ab. Assume Cp of steam = 2.1 kJ/kg K. Calculate dryness fraction of steam. **[NLJIET]**	7	CO2	AN

2	The following information is available from test of a combined separating and throttling calorimeter. (i) Pressure of steam in a steam main = 9.0 bar. (ii) Pressure after throttling = 1.0 bar. (iii) Temperature after throttling = 115 degree centigrade. (iv) Mass of steam condensed after separating = 1.8 kg (v) Mass of water collected in the separator = 0.2 kg. Calculate the dryness fraction of the steam in the main. **[NLJIET]**	**7**	**CO2**	**AN**
3	Determine dryness fraction of steam supplied to a separating and throttling calorimeter. Water separated in separating calorimeter = 0.45 kg. Steam discharge from separating calorimeter = 7 kg. Steam pressure in main pipe = 1.2 MPa. Barometer reading = 760 mm of Hg. Manometer reading = 180 mm of Hg. Temperature of steam after throttling = 140 deg C. Take Cps = 2.1 kJ/kg.K **[NLJIET]**	**7**	**CO2**	**AN**
4	Draw a neat self-explanatory diagram of separating calorimeter. During the test on a separating calorimeter, 2.2 kg moisture was collected and 16 kg of steam left the calorimeter. Calculate dryness fraction of steam. **(Jul-2023) [NLJIET]**	**7**	**CO2**	**AN**
5	Determine dryness fraction of steam supplied to a separating and throttling calorimeter. Water separated in separating calorimeter = 0.45 kg; Steam discharge from throttling calorimeter = 7 kg; Steam pressure in main pipe = 1.2 MPa; Barometer reading= 760 mm of Hg; Manometer reading = 180 mm of Hg; Temperature of steam after throttling = 140 deg C. Take Cps=2.1kJ/kg-K **(Jan-2025) [NLJIET]**	**7**	**CO2**	**AN**

UNIT: 3

HEAT ENGINES AND INTERNAL COMBUSTION ENGINES

Sub Unit 3.1 Heat Engines

Heat engine cycle and Heat engine, working substances, Classification of heat engines, Description and thermal efficiency of Carnot; Rankine; Otto cycle and Diesel cycles.

TOPIC 1 – BASICS OF HEAT ENGINES AND CARNOT CYCLE

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Explain the essential elements of a Heat Engine. **[NLJIET]**	3	CO3	UN
2	Classify Heat engines. **[NLJIET]**	3	CO3	RM
3	Prove that the efficiency of Carnot cycle operating between temperatures T1 and T2 = (T1 – T2) / T1 **[NLJIET]**	3	CO3	AP
4	State limitations of Carnot gas power cycle. **(Jan-2025) [NLJIET]**	3	CO3	RM
5	State the assumptions made in working of Carnot cycle. Also write down the limitations of Carnot cycle. **(Jan-2025-New) [NLJIET]**	3	CO3	RM
6	Define heat engine. Explain the essential requirements of heat engine with suitable figure. **[NLJIET]**	4	CO3	RM
7	Derive the equation of thermal efficiency of Carnot Cycle. Why it cannot be used in practice? Discuss. **[NLJIET]**	4	CO3	AP
8	Derive the equation for efficiency of Carnot cycle. **(Jun-2019) [NLJIET]**	4	CO3	AP
9	Efficiency of Carnot cycle is independent of working fluid, Justify. **(Aug-2022) [NLJIET]**	4	CO3	AP

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Prove that efficiency of Carnot Engine working between temperature limits T1 and T2 is given by the expression $\eta = (T1 - T2) / T1$ **[NLJIET]**	5	CO3	AP
2	Explain Carnot cycle and derive expression for the efficiency of the Carnot cycle. **[NLJIET]**	7	CO3	AP
3	The heat transfer from a heat reservoir is proportional to its temperature: Justify by deriving equation. **(Nov-2020) [NLJIET]**	7	CO3	AP

NUMERICALS				
Sr. No.	**QUESTIONS**	**Marks**	**CO**	**RBT Level**
1	Determine the efficiency of air standard Carnot cycle with the following data. Minimum temperature of the cycle = 27 deg C. Minimum pressure in the cycle = 1 bar. Pressure after isothermal compression = 4.5 bar. Pressure after isentropic compression = 12 bar. Take R = 0.287 kJ/kg-K. Determine also power produced if engine makes 3 cycle/sec. **(Aug-2022) [NLJIET]**	**7**	**CO3**	**AN**
2	It is required to find out the efficiency of an air standard Carnot Cycle with the following data. Minimum temperature of cycle 15 deg C, minimum pressure in cycle 1 bar. Pressure after isothermal compression 3.5 bar. Pressure after isentropic compression 10.5 bar. Assume R = 0.287 kJ/kg.K for air. What power would be produced if engine makes 2 cycles/second. **(Jun-2025-New) [NLJIET]**	**7**	**CO3**	**AN**
3	A Carnot cycle works with isentropic compression ratio of 6 and isothermal expansion ratio of 2. The volume of air at the beginning of isothermal expansion is 0.2 m^3. If T_{max} and p_{max} is limited to 600 K and 20 bar respectively, Determine: 1. Minimum pressure during the cycle, 2. Thermal efficiency of cycle. Show the T-s and p-v diagram. **(Jul-2023) [NLJIET]**	**7**	**CO3**	**AN**

TOPIC 2 – RANKINE CYCLE

SHORT QUESTIONS

Sr. No.	**QUESTIONS**	**Marks**	**CO**	**RBT Level**
1	Discuss Rankine cycle with block diagram. **(Jan-2019) [NLJIET]**	**4**	**CO3**	**AP**
2	Derive an expression for thermal efficiency of Rankine cycle. **(Jan-2025) [NLJIET]**	**4**	**CO3**	**AP**

DESCRIPTIVE QUESTION				
Sr. No.	**QUESTION**	**Marks**	**CO**	**RBT Level**
1	Explain working of Rankine cycle with P-V diagram. Derive the formula for efficiency of Rankine cycle. **[NLJIET]**	**7**	**CO3**	**UN**
NUMERICALS				
Sr. No.	**QUESTIONS**	**Marks**	**CO**	**RBT Level**
1	In ideal Rankine cycle, the steam at inlet to turbine is saturated at a pressure of 35 bar and the exhaust pressure is 0.2 bar. Assume flow rate of 9.5 kg/s. Determine: (1) The pump work, (2) The turbine work, (3) The Rankine efficiency, (4) The dryness at the end of expansion. **[NLJIET]**	**4**	**CO3**	**AN**
2	Consider a steam power plant operating on the simple ideal Rankine cycle. Steam enters the turbine at 3 MPa and 350 deg C and is condensed in the condenser at a pressure of 74 kPa. Draw the T-s diagram and determine the thermal efficiency of this cycle. (Do not neglect the pump work) **(Jan-2024) [NLJIET]**	**7**	**CO3**	**AN**
TOPIC 3 – OTTO CYCLE				
SHORT QUESTIONS				
Sr. No.	**QUESTIONS**	**Marks**	**CO**	**RBT Level**
1	Define air standard efficiency. State any four assumptions considered for analysis of air standard cycle. **[NLJIET]**	**3**	**CO3**	**RM**
2	State assumptions considered for analysis of air standard cycle. **(Jan-2025) [NLJIET]**	**3**	**CO3**	**RM**
3	Derive equation for air standard efficiency of Otto cycle. **(Nov-2020) [NLJIET]**	**4**	**CO3**	**AP**
4	Prove that the efficiency of Otto cycle is greater than that of Diesel cycle for the same compression ratio. **(Jun-2019) [NLJIET]**	**4**	**CO3**	**AP**

DESCRIPTIVE QUESTIONS				
Sr. No.	**QUESTIONS**	**Marks**	**CO**	**RBT Level**
1	With usual notations, derive expression to determine efficiency of Otto cycle. **(Mar-2021) [NLJIET]**	**7**	**CO3**	**AP**
2	Derive, using P-V diagram, the equation to find air standard efficiency of Otto cycle. Use the formula you derived to find which gas from the following will give maximum efficiency for the same compression ratio: exhaust gas ($\gamma = 1.3$), air, and monatomic gas ($\gamma = 1.67$) **(Jan-2024) [NLJIET]**	**7**	**CO3**	**AP**
3	Derive an expression of air standard efficiency of Otto cycle. Draw its P-V and T-S diagram. **(Jan-2025) [NLJIET]**	**7**	**CO3**	**AP**

NUMERICALS				
Sr. No.	**QUESTIONS**	**Marks**	**CO**	**RBT Level**
1	Determine the compression ratio and cycle efficiency of an Otto cycle from following data: Minimum temperature = 25 deg C, Maximum temperature = 1500 deg C, Heat supplied per kg of air = 900 kJ. Take Cv = 0.718 kJ/kg.K and ratio of specific heat $\gamma = 1.4$ **[NLJIET]**	**3**	**CO3**	**AN**
2	In an Otto cycle, the temperature at the beginning and end of the isentropic compression are 316 K and 596 K respectively. Determine the air standard efficiency and compression ratio. Assume $\gamma = 1.4$ **(Mar-2021) [NLJIET]**	**3**	**CO3**	**AN**
3	The engine working on ideal Otto cycle. The temperature at the beginning and at the end of compression is 50 deg C and 400 deg C. Calculate the air standard efficiency and compression ratio. **(Mar-2023) [NLJIET]**	**4**	**CO3**	**AN**
4	Calculate the air standard efficiency of an Otto Cycle engine having a bore of 100 mm and stroke length equal to 150 mm. The clearance volume of cylinder is equal to 250 cm^3. Take $\gamma = 1.4$. **(Jul-2024) [NLJIET]**	**4**	**CO3**	**AN**

5	An engine operation on an air-standard Otto cycle has a compression ratio equal to 7. The conditions at the start of compression are 0.1 MPa and 300 K. The pressure at the end of heat addition is 4 MPa. Determine: (i) thermal efficiency, (ii) net work done per kg of air. Take Cv = 0.718 kJ/kg, and γ = 1.4. **(Jan-2025-New) [NLJIET]**	**4**	**CO3**	**AN**
6	The engine working on ideal Otto cycle. The temperature at the beginning and at the end of compression is 50 deg C and 400 deg C. Calculate the air standard efficiency and compression ratio. **(Jul-2025) [NLJIET]**	**4**	**CO3**	**AN**
7	In air standard Otto Cycle, the Maximum and Minimum temperatures are 1673 K and 288 K. The heat supplied per kg of air is 800 kJ. Calculate. (i) The Compression Ratio. (ii) Efficiency. (iii) Max. & Min. Pressure ratio. Take Cv = 0.718 kJ/kg.K & γ = 1.4 for air. **[NLJIET]**	**7**	**CO3**	**AN**
8	In an Otto Cycle, air at 15 deg C and 1 bar is compressed adiabatically until the pressure is 15 bar. Heat is added at constant volume until pressure rises to 40 bar. Calculate (1) Air standard efficiency (2) Compression ratio and (3) Mean effective pressure for cycle. Assume Cv = 0.718 kJ/kg.K; R = 8.134 kJ/kg.mole.K **[NLJIET]**	**7**	**CO3**	**AN**
9	In an ideal constant volume cycle, the pressure and temperature at the beginning of the compression is 97 kPa and 50 deg C respectively. The volume ratio is 5. The heat supplied during the cycle is 930 kJ/kg of working fluid. Calculate: (1) The maximum temperature attained in the cycle. (2) The thermal efficiency of the cycle. (3) Work done during the cycle/kg of working fluid. **[NLJIET]**	**7**	**CO3**	**AN**
10	An Otto cycle having compression ratio 8 has pressure and temperature at the beginning of compression are 1 bar and 27 deg C respectively. If heat transfer per cycle is 1900 kJ/kg, find pressure and temperature at the end of each process. Take C_V = 0.718 kJ/kg-K. **[NLJIET]**	**7**	**CO3**	**AN**

11	In ideal constant volume cycle, the pressure & temperature at the beginning of compression are 97 kPa & 50 deg C respectively. The volume ratio is 8. The heat is supplied during the cycle is 930 kJ/kg of working fluid. Calculate: 1. The maximum temperature attained in the cycle. 2. The thermal efficiency of cycle. 3. Work done during the cycle/kg of working fluid. **[NLJIET]**	**7**	**CO3**	**AN**
12	An engine working on ideal Otto cycle has a clearance volume of 0.03 m^3 and swept volume of 0.12 m^3. The temperature and pressure at the beginning of compression are 100 deg C and 1 bar respectively. If the pressure at the end of heat addition is 25 bar, Calculate 1. Ideal efficiency of the cycle. 2. Temperature at key points of the cycle. Take $\gamma = 1.4$ for air. **[NLJIET]**	**7**	**CO3**	**AN**
13	A petrol engine has swept volume of 500 cm^3 and clearance volume of 55 cm^3. At suction, pressure and temperature is 1 bar and 30 deg C respectively and maximum temperature in the cycle is 1450 deg C. Calculate air standard efficiency and mean effective pressure of the cycle. Take $\gamma = 1.41$ and R = 0.287 kJ/kg.K **[NLJIET]**	**7**	**CO3**	**AN**
14	In an ideal Otto cycle, the air at the beginning of isentropic compression is at 1 bar and 15 deg C. The ratio of compression is 8. If the heat added during the constant volume process is 1000 kJ/kg, determine (a) The maximum temperature in the cycle (b) The air standard efficiency (c) Work done per kg of air. **(Jun-2019) [NLJIET]**	**7**	**CO3**	**AN**
15	An air standard Otto cycle has compression ratio of 6. The temperature at the start of compression is 25 deg C and pressure is 1 bar. If the maximum temperature of the cycle is 1150 deg C. Calculate (a) The heat supplied and heat rejected per kg of air (b) Net work done per kg of air and (c) Thermal efficiency of the cycle. Assume $\gamma = 1.4$, Cv = 0.778 kJ/kg.K for air. **(Mar-2022) [NLJIET]**	**7**	**CO3**	**AN**

TOPIC 4 – DIESEL CYCLE

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Draw P-V diagram for ideal Diesel cycle and derive expression for its air standard efficiency in terms of temperatures only. **[NLJIET]**	**5**	**CO3**	**AP**
2	Derive equation for efficiency of constant pressure heat addition cycle. **(Mar-2023) [NLJIET]**	**7**	**CO3**	**AP**
3	Derive equation for air standard efficiency of diesel cycle. **(Jan-2025-New) [NLJIET]**	**7**	**CO3**	**AP**
4	With usual notation derive the expression for air standard efficiency of diesel cycle. **(Jun-2025-New) [NLJIET]**	**7**	**CO3**	**AP**
5	Derive an expression of air standard efficiency of Diesel cycle. Draw its P-V and T-S diagram. **(Jul-2025) [NLJIET]**	**7**	**CO3**	**AP**

NUMERICALS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	A diesel engine works on Diesel cycle with a compression ratio of 15 and cut off ratio of 1.75. Calculate the air standard efficiency. Assume $\gamma = 1.4$ **(Jan-2020) [NLJIET]**	**3**	**CO3**	**AN**
2	An engine operating on the ideal diesel cycle has a maximum pressure of 45 bar and a maximum temperature of 1500 deg C. The pressure and temperature of air at the beginning of compression stroke are 1 bar and 27 deg C respectively. Find the air standard efficiency of the cycle. Assume $\gamma = 1.4$ **[NLJIET]**	**4**	**CO3**	**AN**
3	An air standard diesel cycle has compression ratio of 16. The pressure and temperature at the beginning of compression stroke is 1 bar and 20 deg C. The maximum temperature is 1431 deg C. Determine the thermal efficiency and mean effective pressure for this cycle. **[NLJIET]**	**5**	**CO3**	**AN**

4	The compression ratio of an oil engine working on Diesel Cycle is 15. Cut off takes place at 12% of the working stroke. The air is drawn in to cylinder at 100 kPa and 27 deg C. Assume Cp = 1.006 kJ/kg-K and Cv = 0.717 kJ/kg-K. Calculate: 1) Temperature at the end of compression. 2) Pressure at the end of compression. 3) Air std. efficiency of the cycle. **[NLJIET]**	**7**	**CO3**	**AN**
5	In an ideal diesel cycle, the temperature at beginning and at the end of compression are 57 degree centigrade and 603 degree centigrade respectively. The temperatures at beginning and at the end of expansion are 1950 degree centigrade and 870 degree centigrade respectively. Determine the ideal efficiency of the cycle, if pressure at beginning is 1.0 bar. Calculate: maximum pressure in the cycle. **[NLJIET]**	**7**	**CO3**	**AN**
6	An engine operates on the air standard diesel cycle. The conditions at the start of the compression stroke are 353 K and 100 kPa, while at the end of compression stroke the pressure is 4 MPa. The energy absorbed is 700 kJ/kg of air. Calculate 1. The compression ratio 2. The cut-off ratio 3. The work done per kg air 4. The thermal efficiency. **[NLJIET]**	**7**	**CO3**	**AN**
7	The compression ratio of an ideal air standard diesel cycle is 15. Heat supplied at constant pressure is 1470 kJ/kg of air. Show the cycle on p-v diagram and determine the cycle efficiency, if inlet conditions are 300 K and 1 bar. **(Jul-2023) [NLJIET]**	**7**	**CO3**	**AN**
8	Calculate the air standard efficiency of the diesel cycle where the pressure and temperature at the beginning of compression process are 1 bar and 27 deg C respectively. The compression ratio is 14 and the constant pressure heat addition continues up to 7% of the stroke volume. Draw this Diesel cycle on p-v diagram. **(Jul-2024) [NLJIET]**	**7**	**CO3**	**AN**

UNIT: 3

HEAT ENGINES AND INTERNAL COMBUSTION ENGINES

Sub Unit 3.2 Internal Combustion Engines

Introduction, Classification, Engine details, four-stroke/two-stroke cycle Petrol/Diesel engines, Indicated power, Brake Power, Efficiencies.

TOPIC 1 – INTRODUCTION

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Classify I.C. Engines. **(Jan-2020) [NLJIET]**	3	CO3	UN
2	Write the function of following I.C. Engine components: 1. Spark Plug 2. Injector 3. Piston Rings. **(Jul-2024) [NLJIET]**	3	CO3	RM
3	Write the function of following I.C. Engine components: (1) Spark Plug (2) Injector (3) Piston Rings. **(Jun-2025-New) [NLJIET]**	3	CO3	RM
4	Define the following terms: (i) Indicated thermal efficiency. (ii) Compression ratio. (iii) Scavenging. **[NLJIET]**	3	CO3	RM
5	Explain the terms: (i) Swept volume (ii) Clearance volume (iii) Stroke length **(Jan-2019) [NLJIET]**	3	CO3	UN
6	Define the terms: (1) T.D.C. (2) B.D.C. (3) Compression Ratio. **(Jul-2024) [NLJIET]**	3	CO3	RM
7	Classify Internal Combustion Engine. **(Jul-2025) [NLJIET]**	3	CO3	UN
8	Classify the I. C. Engines. **(Jul-2024) [NLJIET]**	4	CO3	UN
9	Give S.I. Units of Followings: (1) Mean effective pressure (2) Efficiency (3) Swept Volume (4) Stroke **[NLJIET]**	4	CO3	RM
10	Give the function of following IC engine parts in one sentence. (i) Piston rings (ii) Connecting rod (iii) Spark plug (iv) Exhaust valve **[NLJIET]**	4	CO3	RM

DESCRIPTIVE QUESTION

Sr. No.	QUESTION	Marks	CO	RBT Level
1	Draw neat sketch of an internal combustion engine, list its parts and state functions of each. **(Mar-2021) [NLJIET]**	7	CO3	UN

TOPIC 2 – FOUR STROKE I.C. ENGINES – CONSTRUCTION, WORKING AND PERFORMANCE

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Give difference between SI and CI engines. **(Jan-2025-New) [NLJIET]**	3	CO3	UN
2	Differentiate between 4-stroke and 2-stroke engine. **[NLJIET]**	4	CO3	UN
3	With neat sketch, explain working of Four stroke petrol engine. **[NLJIET]**	4	CO3	UN
4	Compare 2- stroke engine with 4- stroke engine. **(Jan-2025) [NLJIET]**	4	CO3	UN
5	Write a difference between SI engine and CI engine. **(Jul-2025) [NLJIET]**	4	CO3	UN

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Explain working of Four stroke Diesel Engine with P-V diagram. **[NLJIET]**	5	CO3	UN
2	Write the difference between Two-stroke and Four-stroke cycle I.C. engines. **[NLJIET]**	6	CO3	UN
3	Compare four stroke engine and two stroke engine based on following point/criteria. 1) Number of piston strokes per cycle; 2) Number of crank rotation per cycle; 3) Number of power stroke per min; 4) Power; 5) Flywheel; 6) Size for same power output; 7) Thermal Efficiency and Mechanical Efficiency. **[NLJIET]**	7	CO3	UN
4	Discuss the construction and working of Four stroke Petrol engines. **(Nov-2020) [NLJIET]**	7	CO3	UN
5	With the help of neat sketch, explain the working of Four stroke diesel engine. **[NLJIET]**	7	CO3	UN

6	Write difference between S.I. engine and C.I. engine. **(Mar-2022) [NLJIET]**	7	CO3	UN
7	Write a difference between SI engine and CI engine. **(Jun-2025-New) [NLJIET]**	7	CO3	UN
8	Explain with neat sketch four stoke diesel engine. **(Jan-2025-New) [NLJIET]**	7	CO3	UN

NUMERICALS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	The following data were noted for a 4-cylinder, 4-stroke engine: Diameter = 101 mm, stroke = 114 mm, engine speed = 1600 rpm, fuel consumption = 0.204 kg/min, heating value of fuel = 41,800 kJ/kg, difference in either side of the brake pulley = 378 N, brake pulley radius = 3.35 m. Assume mechanical efficiency = 83%. Calculate: (i) brake thermal efficiency, (ii) indicated thermal efficiency, (iii) fuel consumption per brake power. **(Jan-2025-New) [NLJIET]**	4	CO3	AN
2	A four cylinder four stroke petrol engine has 100 mm bore and stroke is 1.3 times bore. It consumes 4 kg of fuel per hour having calorific value of 40500 kJ/kg. If engine speed is 850 rpm. Find its Indicated thermal efficiency. mep is 0.75 N/mm^2. **[NLJIET]**	5	CO3	AN
3	The following results refer to a test on i.c. engine. Indicated Power = 42 kW; Frictional power = 7 kW; Engine speed = 1800 r.p.m.; Specific fuel consumption per b.p. = 0.30 kg/kWh.; Calorific Value of fuel used = 43000 kJ/kg; Calculate: Mechanical Efficiency, Brake thermal efficiency, Indicated thermal efficiency. **[NLJIET]**	7	CO3	AN
4	The following readings were taken during the test on a single cylinder four stroke Oil engine: Cylinder diameter = 270 mm. Stroke Length = 380 mm. Mean effective pressure = 6 bar. Engine Speed = 250 rpm. Net load on brake = 1000 N. Effective mean diameter of brake = 1.5 m. Fuel used = 10 kg/hr. C.V. of Fuel = 44400 kJ/kg. Calculate**: -** Brake Power. Indicated	7	CO3	AN

	Power. Mechanical Efficiency. Indicated Thermal Efficiency. **(Jan-2019) [NLJIET]**			
5	The following readings were taken during the test on a single cylinder four stroke Oil engine: Cylinder diameter = 270 mm. Stroke Length = 380 mm. Mean effective pressure = 6 bar. Engine Speed = 250 rpm. Net load on brake = 1000 N. Effective mean diameter of brake = 1.5 m. Fuel used = 10 kg/hr. C.V. of Fuel = 44400 kJ/kg. Calculate**: -** 1) Brake Power. 2) Indicated Power. 3) Indicated Thermal Efficiency. **(Mar-2023) [NLJIET]**	**7**	**CO3**	**AN**
6	Following readings were taken during test of single cylinder four stroke oil engine. Cylinder diameter = 250 mm**;** Stroke length = 400 mm**;** Mean effective pressure = 7 bar**;** Engine speed = 250 r.p.m.; Net load on brake = 1080 Newton**;** Effective diameter of brake = 1.5 meter**;** Fuel used per hour = 10 kg**;** Calorific value of fuel = 44300 kJ/kg**.** Calculate (1) Indicated power, (2) Brake power, (3) Mechanical efficiency and (4) Indicated thermal efficiency. **[NLJIET]**	**7**	**CO3**	**AN**
7	The following readings were taken during the test of four stroke single cylinder petrol engine: Load on the brake drum = 50 kg. Diameter of brake drum = 1250 mm. Spring balance reading = 7 kg. Engine speed = 450 rpm. Fuel consumption = 4 kg/hr Calorific value of the fuel = 43000 kJ/kg. Calculate: (i) Indicated thermal efficiency. (ii) Brake thermal efficiency. Assume mechanical efficiency as 70%. **[NLJIET]**	**7**	**CO3**	**AN**
8	The following results refer to a test on C.I. engine. Indicated power = 37 kW. Frictional power = 06 kW. Brake specific fuel consumption = 0.28 kg/kWh. Calorific value of fuel = 44300 kJ/kg. Calculate: (i) Mechanical efficiency (ii) Brake thermal efficiency (iii) Indicated thermal efficiency **[NLJIET]**	**7**	**CO3**	**AN**

9	A two cylinder four stroke petrol engine has swept volume of 1.1×10^{-3} m^3. It runs at 950 rpm and consumes 2.2 kg of petrol per hour having C.V. of 43000 kJ/kg. The mean effective pressure in both cylinders is 7.5 bars. Determine indicated thermal efficiency and relative efficiency if clearance volume is 15% of swept volume. **(Aug-2022) [NLJIET]**	**7**	**CO3**	**AN**
10	A petrol engine with a stroke length of 200 mm and diameter of 150 mm has a clearance volume of 7×10^5 mm^3. If the indicated thermal efficiency is 0.30, find the relative efficiency. If the effective pressure is 5 bar and engine runs at 1000 rpm. Find the IP of the engine. Take $\gamma = 1.4$ **(Nov-2020) [NLJIET]**	**7**	**CO3**	**AN**
11	A 4 cylinder 4-stroke marine oil engine has cylinder diameter of 490 mm and a piston stroke of 1000 mm. The engine uses 130 kg of fuel of calorific value 42,000 kJ/kg in one hour when running at 2 rev/s. The torque transmitted at the engine coupling is 22 kN.m and indicated mean effective pressure 710 kN/m^2. Determine (i) Indicated power (ii) Brake power (iii) Indicated thermal efficiency (iv) Brake thermal efficiency (v) Mechanical efficiency. **(Aug-2022) [NLJIET]**	**7**	**CO3**	**AN**
12	During a trial on single cylinder oil engine, working on the four stroke cycle and fitted with rope brake the following readings were taken - Effective diameter of brake wheel = 630 mm, dead load on the brake = 200 N, spring balance reading = 30 N, Speed = 450 rpm, Area of indicator diagram = 420 mm^2, length of indicator diagram = 60 mm, spring scale = 1.1 bar per mm, diameter of cylinder = 100 mm, stroke = 150 mm, quantity of oil used = 0.815 kg/hr, calorific value of fuel = 42,000 kJ/kg, Calculate brake power, indicated power, mechanical efficiency, brake thermal efficiency and brake specific fuel consumption. **(Mar-2021) [NLJIET]**	**7**	**CO3**	**AN**

13	Ford car having a four cylinder, four stroke petrol engine has 100 mm bore and stroke is 1.25 times the bore. It consumes 4 kg of fuel per hour having calorific value of 41,000 kJ/kg. The engine speed is 800 rpm. Calculate indicated thermal efficiency if mean effective pressure is 0.75 MPa. **(Mar-2022) [NLJIET]**	**7**	**CO3**	**AN**
14	Two cylinder four stroke diesel engine has total swept volume of 870 cm^3. Following data is available with test on the engine. Engine speed = 300 rpm; Brake torque= 50 N-m; pmi = 10 bar; Calculate (1) Indicated power (2) Brake power (3) Mechanical efficiency. **(Mar-2023) [NLJIET]**	**7**	**CO3**	**AN**
15	Indicated power of a 6-cylinder 4-stroke engine is 150 kW at an average piston speed of 300 m/min. Stroke to bore ratio is 1.25. If mean effective pressure is 650 kN/m^2, determine crankshaft speed. What is a flywheel and what is its function in IC engine? **(Jul-2023) [NLJIET]**	**7**	**CO3**	**AN**
16	A 4-cylinder 4-stroke petrol engine has a bore 60 mm and a stroke of 90 mm. The rated speed is 2800 rpm and torque is 55 N-m. Fuel consumption is 6.75 litre/hr. Specific gravity of petrol is 0.76 and calorific value is 44200 kJ/kg. Calculate brake power, brake mean effective pressure, brake thermal efficiency and brake specific fuel consumption. **(Jul-2023) [NLJIET]**	**7**	**CO3**	**AN**
17	A 4-stroke 6-cylinder IC engine has a stroke volume of 1.75 liters and is operating at a mean effective pressure of 6 bar. At what crankshaft rpm will the engine develop 35 hp? (Take 1 hp = 736 watts). What is the function of a carburetor and a fuel injector in an IC engine? **(Jan-2024) [NLJIET]**	**7**	**CO3**	**AN**
18	A petrol engine has a compression ratio of 6 and develops 15 kW. Its brake thermal efficiency is half of its air standard efficiency. Find the fuel consumption. Take calorific value of petrol as 41500 kJ/kg. Why are more two-wheeler automobiles made with a 4-stroke engine instead of a 2-stroke engine? Give three technical reasons for this. **(Jan-2024) [NLJIET]**	**7**	**CO3**	**AN**

19	A Diesel engine develops 5 kW brake power at a speed of 1000 rpm. The mechanical efficiency of the engine is 75% and indicated thermal efficiency is equal to 30%. The diesel fuel having specific gravity = 0.87 and calorific value = 42700 kJ/kg is used. Calculate: 1. Fuel consumption in kg/h. 2. Brake thermal efficiency 3. Brake specific fuel consumption. **(Jul-2024) [NLJIET]**	**7**	**CO3**	**AN**
20	A six-cylinder four stroke petrol engine develop 300 kW brake power at 2500 rpm. The stroke to bore ratio is 1.25. Assuming the mechanical efficiency as 80% and mean effective pressure of 9 bar, determine the bore and stroke of engine. Also find the fuel consumption in kg/hr if indicated thermal efficiency is 30% and CV of fuel used is 41900 kJ/kg. **(Jun-2025-New) [NLJIET]**	**7**	**CO3**	**AN**
21	Following observations were recorded during a test on a single cylinder oil engine. Bore = 300 mm; Stroke = 450 mm; Speed = 300 r.p.m.; i.m.e.p. = 6 bar; Net brake load = 1.5 kN; Brake drum diameter = 1.8 m; Brake rope diameter = 2 cm. Calculate (i) I.P. (ii) B.P. (iii) Mechanical efficiency. **[NLJIET]**	**8**	**CO3**	**AN**

TOPIC 3 – TWO STROKE I.C. ENGINES – CONSTRUCTION, WORKING AND PERFORMANCE

SHORT QUESTION

Sr. No.	QUESTION	Marks	CO	RBT Level
1	Explain with the help of neat sketches, the working of Two stroke petrol engine. **(Mar-2023) [NLJIET]**	**4**	**CO3**	**UN**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Explain with neat sketch, construction and working of Two stroke petrol engine. **[NLJIET]**	**7**	**CO3**	**UN**
2	With neat sketch, describe the working of Two stroke diesel engine. **[NLJIET]**	**7**	**CO3**	**UN**

NUMERICALS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Determine the brake thermal efficiency of an engine with following data: Brake Power = 80 kW; Fuel consumption rate = 20 kg/hour; Calorific value of the fuel = 43 MJ/kg. **(Jan-2020) [NLJIET]**	**4**	**CO3**	**AN**
2	During testing of single cylinder two stroke petrol engine following data is obtained - Brake torque 640 Nm, Cylinder diameter 21 cm, speed 350 rpm, stroke 28 cm, mep 5.6 bar, oil consumption 8.16 kg/hr, C.V. 42705 kJ/kg. Determine Mechanical efficiency. Indicated thermal Efficiency. Brake thermal efficiency. Brake specific fuel consumption. **[NLJIET]**	**7**	**CO3**	**AN**
3	A four cylinder two stroke petrol engine with stroke to bore ratio 1.2 develops 35 kW brake power at 2200 rpm. The mean effective pressure in each cylinder is 9 bar and mechanical efficiency is 78 %. Determine (1) Diameter and Stroke of each cylinder (2) Brake thermal efficiency (3) Indicated thermal efficiency; If fuel consumption is 8 kg/hr having C.V. = 43000 kJ/kg. **[NLJIET]**	**7**	**CO3**	**AN**
4	During testing of single cylinder two stroke petrol engine, following data is obtained: Brake torque 640 Nm, cylinder diameter 21 cm, Speed 350 rpm, stroke 28 cm, mean effective pressure 5.6 bar, oil consumption 8.16 kg/hr, C.V. = 42705 kJ/kg. Find, (1) Mechanical efficiency (2) Indicated thermal efficiency (3) Brake thermal efficiency (4) Brake specific fuel consumption. **[NLJIET]**	**7**	**CO3**	**AN**
5	The following readings were observed during test on Two Stroke Single Cylinder Diesel engine: Bore = 22 cm; Stroke = 28 cm; Speed = 350 rpm; Net brake load = 65 kg; Effective brake drum diameter = 100 cm; Mean Effective Pressure = 3 bar; Fuel consumption = 4 kg/hr; C.V. of fuel = 43 MJ/kg; Calculate: (1) Indicated Power (2) Brake Power (3) Mechanical Efficiency (4) Brake thermal Efficiency. **(Jan-2020) [NLJIET]**	**7**	**CO3**	**AN**

6	A 4-cylinder, two stroke cycle petrol engine develops 30 kW at 2500 rpm. The mean effective pressure on each piston is 8 bar and mechanical efficiency is 80%. Calculate the diameter and stroke of each cylinder if stroke to bore ratio is 1.5. Also the fuel consumption of engine, if brake thermal efficiency is 28% and calorific value is 43900 kJ/kg. **(Mar-2022) [NLJIET]**	**7**	**CO3**	**AN**
7	The following data is available for 2-stroke diesel engine: Bore = 10 cm, stroke = 15 cm, engine speed=1000 rpm, torque developed= 58 N-m, ηm= 80%, ηith= 40%, calorific value of fuel = 44000 kJ/kg. Find: (i) Indicated power (ii) Mean effective pressure (iii) Brake specific fuel consumption. **(Jan-2025) [NLJIET]**	**7**	**CO3**	**AN**

UNIT: 4
STEAM BOILERS

Introduction, Classification, Cochran, Lancashire and Babcock and Wilcox boiler, Functioning of different mountings and accessories.

TOPIC 1 – INTRODUCTION AND TYPES OF STEAM BOILERS
SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	How do you classify steam boilers? **(Mar-2022) [NLJIET]**	**3**	**CO4**	**RM**
2	How do you classify steam boilers? **(Jul-2025) [NLJIET]**	**3**	**CO4**	**RM**
3	Classify boilers according to (i) relative position of hot gases and water, (ii) axis of the shell, (iii) method of water circulation. Give examples in each. **(Jul-2023) [NLJIET]**	**3**	**CO4**	**RM**
4	Define boiler. Also mention at least five essential requirements of a good boiler. **(Jan-2025-New) [NLJIET]**	**3**	**CO4**	**RM**
5	What is boiler? Compare fire tube boiler with water tube boiler. **(Jan-2020) [NLJIET]**	**4**	**CO4**	**UN**
6	Explain difference between fire tube boiler and water tube boiler. **(Mar-2023) [NLJIET]**	**4**	**CO4**	**UN**

7	Draw labelled diagram of Babcock and Wilcox boiler. **(Mar-2022) [NLJIET]**	4	CO4	RM
8	Draw labelled diagram of Babcock and Wilcox boiler. **(Jul-2025) [NLJIET]**	4	CO4	RM
9	Explain Equivalent evaporation and factor of evaporation. **(Nov-2020) [NLJIET]**	4	CO4	UN
10	Discuss construction & working of Cochran boiler with sketch. **[NLJIET]**	4	CO4	UN
11	Explain: Smoke tube internally fired horizontal type stationary boiler. **(Jan-2019) [NLJIET]**	4	CO4	UN
12	Classify the Boilers. **(Jul-2024) [NLJIET]**	4	CO4	RM
13	List essential qualities of a good boiler. **(Mar-2021) [NLJIET]**	4	CO4	RM
14	List essential qualities of a good boiler. **(Jun-2025-New) [NLJIET]**	4	CO4	RM
15	What are the advantages and disadvantages of water tube boiler over fire tube boiler? **(Aug-2022) [NLJIET]**	4	CO4	RM

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Sketch the Cochran boiler and label all important mountings and accessories. **[NLJIET]**	5	CO4	UN
2	Explain construction and working of Babcock and Wilcox boiler with line diagram. **(Aug-2022) [NLJIET]**	7	CO4	UN
3	Explain Cochran boiler with neat sketch & give its advantages and disadvantages. **[NLJIET]**	7	CO4	UN
4	What is boiler? Discuss construction & working of Cochran boiler with neat sketch. **[NLJIET]**	7	CO4	UN
5	Discuss the construction details and working of Cochran boiler with neat sketch. **(Jan-2020) [NLJIET]**	7	CO4	UN
6	Explain with neat sketch - Cochran Boiler. **(Mar-2023) [NLJIET]**	7	CO4	UN

7	Explain construction and working of Lancashire boiler. **[NLJIET]**	**7**	**CO4**	**UN**
8	Explain the characteristics of a Lancashire boiler with neat diagrams. **(Jul-2024) [NLJIET]**	**7**	**CO4**	**UN**
9	Differentiate between fire tube and water tube boiler. Explain Babcock and Wilcox Boiler construction with neat sketch. **[NLJIET]**	**7**	**CO4**	**UN**
10	Explain with neat sketch working of Cochran boiler. **(Jan-2025) [NLJIET]**	**7**	**CO4**	**UN**
11	Explain with neat sketch working of Babcock and Wilcox boiler. **(Jan-2025-New) [NLJIET]**	**7**	**CO4**	**UN**
12	Explain with neat sketch Cochran boiler. **(Jul-2025) [NLJIET]**	**7**	**CO4**	**UN**

TOPIC 2 – BOILER MOUNTINGS AND ACCESSORIES

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	List different mountings of boiler and explain any one in brief. **(Nov-2020) [NLJIET]**	**3**	**CO4**	**UN**
2	What is an economizer? Why is it used in boiler plant? **(Jul-2024) [NLJIET]**	**3**	**CO4**	**RM**
3	Explain very briefly the function of following mountings: (i) Steam stop valve (ii) Feed check valve (iii) Blow-off cock (iv) Water level indicator (v) Pressure gauge (vi) Safety valve. **[NLJIET]**	**3**	**CO4**	**RM**
4	With neat sketch, explain construction and working of Pressure gauge. **[NLJIET]**	**3**	**CO4**	**RM**
5	State the function of the following: (1) Pressure gauge (2) Fusible plug (3) Superheater **[NLJIET]**	**3**	**CO4**	**RM**
6	State the function of the following - (1) Fusible plug (2) Economizer (3) Safety valve **[NLJIET]**	**3**	**CO4**	**RM**
7	Give function of following: (i) Water level indicator (ii) Fusible plug (iii) Economizer **[NLJIET]**	**3**	**CO4**	**RM**

8	State the function of the following in boilers: Economizer, Air preheater, Superheater. **(Jan-2024) [NLJIET]**	**3**	**CO4**	**RM**
9	What do you mean by 'boiler mountings' and 'boiler accessories'? Write three examples of them. **(Jan-2024) [NLJIET]**	**3**	**CO4**	**RM**
10	Explain the construction and function of Steam Trap with neat sketch. Also mention its specific location in the system. **(Jun-2019) [NLJIET]**	**3**	**CO4**	**UN**
11	Explain the construction and function of Steam Trap with neat sketch. **(Nov-2020) [NLJIET]**	**3**	**CO4**	**UN**
12	Explain the construction and function of Steam separator with neat sketch. Also mention its specific location in the system. **(Jun-2019) [NLJIET]**	**3**	**CO4**	**UN**
13	Economizer used to increase efficiency of boiler. Justify this statement. **(Aug-2022) [NLJIET]**	**3**	**CO4**	**UN**
14	State the function of the following in boilers: Feed check valve, Blow off cock, Fusible plug. **(Jul-2023) [NLJIET]**	**3**	**CO4**	**RM**
15	Explain Fusible plug with neat sketch. **[NLJIET]**	**4**	**CO4**	**UN**
16	Explain function and working of Fusible plug used in the boiler. **[NLJIET]**	**4**	**CO4**	**UN**
17	Explain with neat sketch, the constructional details and working of the Ramsbottom type spring loaded Safety Valve. **[NLJIET]**	**4**	**CO4**	**UN**
18	Explain Green's economizer with neat sketch. **[NLJIET]**	**4**	**CO4**	**UN**
19	Explain with neat sketch, Bourdon tube type pressure gauge. **[NLJIET]**	**4**	**CO4**	**UN**
20	Differentiate between boiler mountings and accessories. **(Jan-2020) [NLJIET]**	**4**	**CO4**	**UN**
21	Discuss with neat sketch, any two boiler accessories. **(Nov-2020) [NLJIET]**	**4**	**CO4**	**UN**
22	With neat sketch, explain construction and working of Water level indicator. **(Mar-2021) [NLJIET]**	**4**	**CO4**	**UN**
23	Discuss briefly location and function of the followings; Water level indicator; Pressure Gauge. **(Jan-2025) [NLJIET]**	**4**	**CO4**	**UN**

24	Differentiate between boiler mountings and accessories. **(Jan-2025) [NLJIET]**	**4**	**CO4**	**UN**
25	List different mountings of boiler and explain any one in brief. **(Jun-2025-New) [NLJIET]**	**4**	**CO4**	**UN**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Explain Green's economizer with neat sketch. **[NLJIET]**	**6**	**CO4**	**UN**
2	Explain "Economizer" as an accessory of boiler with neat sketch. Also give its advantages and disadvantages. **[NLJIET]**	**7**	**CO4**	**UN**
3	Explain Economizer and Air-preheater with neat sketch. **[NLJIET]**	**7**	**CO4**	**UN**
4	Explain with neat sketch, the construction and working of (i) Fusible plug and (ii) Air preheater. **(Jan-2020) [NLJIET]**	**7**	**CO4**	**UN**
5	With neat sketch, explain construction and working of Super heater. **[NLJIET]**	**7**	**CO4**	**UN**
6	State the function of Fusible Plug, Economiser, Safety valves, Water level indicator, Superheater, Pressure gauge, Air pre-heater. **[NLJIET]**	**7**	**CO4**	**RM**
7	With neat sketch, explain the construction and working of (1) Fusible plug. (2) Pressure gauge. **[NLJIET]**	**7**	**CO4**	**UN**
8	What do you mean by mounting and accessories? Enlist different boiler mountings and accessories with their functions. **(Jan-2025-New) [NLJIET]**	**7**	**CO4**	**UN**

UNIT: 5

AIR COMPRESSORS AND PUMPS

Sub Unit 5.1 Air Compressors

Types and operation of Reciprocating and Rotary air compressors, significance of Multistage.

TOPIC 1 – INTRODUCTION AND RECIPROCATING AIR COMPRESSOR

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	List out applications of compressed air. **(Mar-2022) [NLJIET]**	**3**	**CO4**	**RM**

2	Classify Air Compressors. **(Jan-2020) [NLJIET]**	3	CO4	UN
3	Draw a neat sketch of p-v diagram for single stage compressor with clearance. **(Jun-2019) [NLJIET]**	3	CO4	RM
4	Draw a neat sketch of p-v diagram showing Free Air Delivery for air compressor. **(Nov-2020) [NLJIET]**	3	CO4	RM
5	Define Compressor. State uses of compressed air. **(Jan-2025) [NLJIET]**	3	CO4	RM
6	Define following with reference to air compressor; (i) Free air delivery (ii) Swept volume (iii) Volumetric efficiency. **(Jan-2025) [NLJIET]**	3	CO4	RM
7	What are the uses of compressed air? **(Mar-2023) [NLJIET]**	4	CO4	RM
8	Give classification of air compressors. **(Jan-2024) [NLJIET]**	4	CO4	UN
9	List out applications of compressed air. **(Jul-2025) [NLJIET]**	4	CO4	RM
10	Explain need of multi staging in reciprocating air compressor with its advantages. **(Jan-2019) [NLJIET]**	4	CO4	UN
11	Explain the following terms: Volumetric efficiency, Compression ratio. **(Mar-2021) [NLJIET]**	4	CO4	UN
12	With usual notations, prove that volumetric efficiency of reciprocating air compressor is $1–C\ [(P2\ /\ P1)^{1/n} – 1]$, where C = clearance volume ratio. **[NLJIET]**	4	CO4	AP
13	Prove that the work done per kg of air in Reciprocating Air Compressor neglecting clearance volume is given by $W=R*T1*n/(n-1)[(Rp)^{(n-1)/n}-1]$, Where Rp = Pressure Ratio. **[NLJIET]**	4	CO4	AP
14	Outline the technical meaning of following terms used in air compressor: single acting compressor, single stage compressor, pressure ratio, swept volume, volumetric efficiency. **(Jul-2023) [NLJIET]**	4	CO4	RM
15	Define with regard to compressors: mean effective pressure, indicated power, brake power, mechanical efficiency. **(Jul-2023) [NLJIET]**	4	CO4	RM

16	Define with regard to compressors: double-acting compressor, multistage compressor, displacement (swept) volume, pressure ratio. **(Jan-2024) [NLJIET]**	**4**	**CO4**	**RM**
17	Explain with P-V diagram need of multi staging in reciprocating air compressor. **(Jan-2025-New) [NLJIET]**	**4**	**CO4**	**UN**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Why multistage compression has more benefit over single stage compression for achieving high pressure ratio? Explain with P-V diagram. **[NLJIET]**	**5**	**CO4**	**UN**
2	Derive an expression for compressor without clearance $W = P * V * \log_e(P2/P1)$ for isothermal compression. **[NLJIET]**	**6**	**CO4**	**AP**
3	What are the applications of compressor? Derive an expression of work done for single stage single acting reciprocating air compressor without clearance. **[NLJIET]**	**7**	**CO4**	**AP**
4	Classify Air Compressors. Give the uses or application of compressed air. **(Jun-2015) [NLJIET]**	**7**	**CO4**	**UN**
5	Explain the concept of Multi staging in reciprocating compressor. What are its advantages? **[NLJIET]**	**7**	**CO4**	**UN**
6	Define volumetric efficiency with p-v diagram and usual notations. Prove that volumetric efficiency of reciprocating compressor is $1-C[(p2/p1)^{1/n} -1]$ **[NLJIET]**	**7**	**CO4**	**AP**
7	Derive the expression of work done by a single stage single acting reciprocating air compressor neglecting clearance volume. Consider adiabatic process for expansion of air. **[NLJIET]**	**7**	**CO4**	**AP**
8	With usual notations, derive an expression for work done for single stage single acting reciprocating air compressor by considering clearance volume. **[NLJIET]**	**7**	**CO4**	**AP**
9	Derive equation of work required for a single stage single acting reciprocating air compressor undergoing polytropic compression considering clearance volume. **[NLJIET]**	**7**	**CO4**	**AP**

NUMERICALS				
Sr. No.	**QUESTIONS**	**Marks**	**CO**	**RBT Level**
1	A single stage, single acting compressor has a bore of 170 mm and stroke of 260 mm. It runs at 130 rpm. The suction pressure is 1 bar and delivery pressure is 9 bar. Find the indicated power if compression 1. Follows the law $pv^{1.25}$ = constant and 2. Isothermal. Also find isothermal efficiency. Assume there is no clearance volume. **[NLJIET]**	**5**	**CO4**	**AN**
2	A single cylinder, single acting air compressor has a cylinder diameter of 150 mm and stroke of 300 mm. It draws air into its cylinder at a pressure of 1 bar and temperature 27 deg C. This air is then compressed adiabatically to a pressure of 8 bar if the compressor runs at a speed of 120 rpm. Find (i) Mass of the air compressed per cycle (ii) Work required per cycle (iii) Power required to drive the compressor. Neglect the clearance volume and take R = 0.287 kJ/kg.K **[NLJIET]**	**6**	**CO4**	**AN**
3	A single stage, single cylinder reciprocating air compressor with negligible clearance takes 1 m^3 of air per minute at 1.013 bar and 15 deg C. The delivery pressure is 7 bar. Assuming law of compression $pv^{1.35}$ = C, R = 0.287 kJ/kg.K Calculate: (1) Mass of air delivered per minute (2) Delivery temperature (3) Indicated Power (4) Isothermal efficiency. **[NLJIET]**	**7**	**CO4**	**AN**
4	A single stage reciprocating compressor takes 1 m^3 of air permissible at 1.013 bar and 15 deg C and delivers it at 7 bar. Assuming that the law of compression is $PV^{1.35}$ = Constant and the clearance is negligible. Calculate the Indicated Power. **[NLJIET]**	**7**	**CO4**	**AN**
5	Air is to be compressed in a single stage reciprocating compressor from 1.013 bar and 15 deg C to 7 bar. Calculate the indicated power required for free air delivery of 0.3 m^3/min when the compression process is 1. Isentropic 2. Reversible isothermal 3. Polytropic with n = 1.25. **(Aug-2022) [NLJIET]**	**7**	**CO4**	**AN**

6	A single stage air compressor is required to compress 94 m^3 of air/min from 1 bar and 25 deg C to 9 bar. Find the temperature at the end of the compression, work done, power required and heat rejected during each of the following processes: 1) Isothermal 2) Adiabatic 3) Polytropic following the law $pV^{1.25}$ = C. Assume no clearance. **[NLJIET]**	**7**	**CO4**	**AN**
7	A single stage single acting air compressor has intake pressure 1 bar and delivery pressure 12 bar. The compression and expansion follows the law $pV^{1.3}$=constant. The piston speed and rotations of shaft is 180 m/min and 350 rpm respectively. Indicated power is 30 kW and volumetric efficiency is 92%. Determine bore and stroke. **[NLJIET]**	**7**	**CO4**	**AN**
8	What is air compressor? Air is to be compressed through a pressure ratio of 10 from a pressure of 1 bar in a single stage air compressor. Free air delivery is 3 m^3/min. Swept volume = 14 liters. Index of compression is 1.3. Neglect clearance volume. Calculate (1) Power required in kW (2) Rotational speed of the compressor. **[NLJIET]**	**7**	**CO4**	**AN**
9	A single stage air compressor is required to compress 72 m^3 of air per minute from 15 deg C and 1 bar to 8 bar pressure. Find the temperature at the end of the compression, work done, power and heat rejected during each of the following processes: 1. Isothermal compression 2. Polytropic compression following the law $pV^{1.25}$ = C. Neglect Clearance. **[NLJIET]**	**7**	**CO4**	**AN**
10	Calculate the dimensions (diameter and stroke) of a double acting reciprocating compressor neglecting clearance volume from the following additional data: compressor power = 10 kW, Compressor ratio = 7, Inlet pressure=1 bar, Piston speed = 180 m/min, Law of compression = $PV^{1.3}$= C, L/D = 1.5. **(Jun-2025-New) [NLJIET]**	**7**	**CO4**	**AN**

TOPIC 2 – ROTARY, CENTRIFUGAL AND AXIAL FLOW COMPRESSOR

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Compare reciprocating compressor and rotary compressor. **(Mar-2021) [NLJIET]**	3	CO4	UN
2	Explain with neat diagram, the working of Centrifugal compressor. **(Jul-2024) [NLJIET]**	3	CO4	UN
3	What is compressor? Give the classification of Rotary Compressors. **[NLJIET]**	4	CO4	RM
4	What are the differences between Reciprocating and Rotary compressor? **(Aug-2022) [NLJIET]**	4	CO4	UN
5	Elaborate differences between reciprocating and rotary compressor? **(Jun-2025-New) [NLJIET]**	4	CO4	UN
6	Explain with neat diagram, the construction and working of Roots Blower. **(Jul-2024) [NLJIET]**	4	CO4	UN

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Explain difference between Reciprocating and Rotodynamic compressor. **[NLJIET]**	5	CO4	UN
2	Classify Rotary air compressors. Explain the construction and working of Centrifugal compressor with neat sketch. **[NLJIET]**	7	CO4	RM
3	Distinguish between Reciprocating and Rotary Compressor. **(Jan-2019) [NLJIET]**	7	CO4	UN
4	Explain construction and working of Centrifugal compressor with neat sketch. **(Mar-2022) [NLJIET]**	7	CO4	UN
5	Classify the air compressor. Differentiate between reciprocating compressor and rotary compressor. **[NLJIET]**	7	CO4	UN

UNIT: 5

AIR COMPRESSORS AND PUMPS

Sub Unit 5.2 Pumps

Types and operation of Reciprocating, Rotary and Centrifugal pumps, Priming.

TOPIC 1 – INTRODUCTION AND RECIPROCATING PUMP

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	What do you understand by word pump? Draw neat sketch of Single acting reciprocating pump with nomenclature. **[NLJIET]**	3	CO4	UN
2	Discuss with neat sketch, Diaphragm pump. **(Nov-2020) [NLJIET]**	3	CO4	UN
3	Explain with neat sketch, Single acting Plunger type pump. **(Jan-2019) [NLJIET]**	3	CO4	UN
4	Classify pumps on basis of principle, construction, and fluid flow direction in pump. **(Mar-2021) [NLJIET]**	3	CO4	RM
5	Explain single acting reciprocating pump. **(Jul-2025) [NLJIET]**	3	CO4	UN
6	Explain working of Single acting reciprocating pump with air vessels. **[NLJIET]**	4	CO4	UN
7	Explain Double acting reciprocating pump with a neat sketch. **[NLJIET]**	4	CO4	UN
8	Explain with sketch, the working of Single acting piston pump. **[NLJIET]**	4	CO4	UN
9	Explain - Single acting reciprocating pump. **(Mar-2022) [NLJIET]**	4	CO4	UN
10	Discuss with neat sketch, Diaphragm pump. **(Jan-2019) [NLJIET]**	4	CO4	UN
11	Classify the pumps. **(Jul-2024) [NLJIET]**	4	CO4	RM
12	Draw a schematic diagram of reciprocating pump. Show the variation of discharge of water with crank angle for single-acting reciprocating pump. **(Jul-2023) [NLJIET]**	4	CO4	RM

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Explain working of Double acting reciprocating pump and Bucket pump with neat sketch. **[NLJIET]**	5	CO4	UN
2	Explain following terms associated with pumps – 1. Priming in Pumps 2. Head 3. Air chamber **[NLJIET]**	6	CO4	UN
3	What is the function of pump? Classify the pumps. Explain with sketch, the working of Single acting piston pump. **[NLJIET]**	7	CO4	UN

NUMERICAL

Sr. No.	QUESTION	Marks	CO	RBT Level
1	A double acting reciprocating pump has piston diameter of 150 mm and stroke length of 225 mm. The suction and delivery heads are 4 m and 12 m respectively. If the speed of the pump is 80 rpm and the actual quantity of water discharged is 0.61 m^3/min. Find the percentage slip, the coefficient of discharge and the power required to drive the pump, if the efficiency of the pump is 80%. **(Jun-2019) [NLJIET]**	7	CO4	AN

TOPIC 2 – CENTRIFUGAL PUMP AND PRIMING

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Explain the working principle of Centrifugal pump with neat sketch? **[NLJIET]**	3	CO4	UN
2	What is priming? Why priming is required in centrifugal pump but not in reciprocating pumps? **(Mar-2022) [NLJIET]**	3	CO4	UN
3	What is priming? Why is it required in centrifugal pump? **(Jan-2025-New) [NLJIET]**	3	CO4	UN
4	What is priming? Why priming is required in centrifugal pump? **(Jun-2025-New) [NLJIET]**	3	CO4	UN

5	Explain construction and working of Centrifugal pump with sketch. **(Jan-2020) [NLJIET]**	4	CO4	UN
6	Draw a labeled figure of centrifugal pump indicating all main components. **(Jul-2023) [NLJIET]**	4	CO4	RM
7	Explain with neat diagram, the working of centrifugal pump. **(Jul-2024) [NLJIET]**	4	CO4	UN
8	What is priming? Why priming is required in centrifugal pump but not in reciprocating pump? **(Aug-2022) [NLJIET]**	4	CO4	UN
9	Compare centrifugal pumps v/s reciprocating pumps. **(Jan-2024) [NLJIET]**	4	CO4	UN
10	Classify the Centrifugal pump and explain with neat sketch the Vortex type centrifugal pump. **[NLJIET]**	4	CO4	UN
11	Give comparison between reciprocating and centrifugal pump. **(Jan-2025-New) [NLJIET]**	4	CO4	UN

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Explain working of main parts of Centrifugal pump with neat sketch. **[NLJIET]**	5	CO4	UN
2	Explain construction and working of Centrifugal pump with sketch. **[NLJIET]**	6	CO4	UN
3	Explain priming and priming methods. **[NLJIET]**	7	CO4	UN
4	State the different types of Centrifugal pumps. Describe Diffuser type of centrifugal pump. **[NLJIET]**	7	CO4	UN
5	What is the function of a pump? Explain with neat sketch, working of Centrifugal pump. **[NLJIET]**	7	CO4	UN
6	Classify Centrifugal pumps. With neat sketch, explain the function of each part of Centrifugal pump. **[NLJIET]**	7	CO4	UN
7	Explain construction and working of Centrifugal pump with a neat sketch. **(Nov-2020) [NLJIET]**	7	CO4	UN
8	What is priming? Explain with neat sketch, centrifugal pump. **(Mar-2023) [NLJIET]**	7	CO4	UN

9	What is priming? Explain with neat sketch centrifugal pump. **(Jul-2025) [NLJIET]**	7	CO4	UN
10	Explain with neat sketch working of centrifugal pump. **(Jan-2025) [NLJIET]**	7	CO4	UN
NUMERICAL				
Sr. No.	**QUESTION**	**Marks**	**CO**	**RBT Level**
1	A centrifugal pump handles water and drives by motor which consumes 32 kW. The pump running at 2000 rpm. The motor efficiency is 92%. The height of pump axis from sump water surface is 6 m and produces a delivery head of 24 m. The discharge rate of water is 260 m^3/hr. Calculate the efficiency of pump. **[NLJIET]**	4	CO4	AN
TOPIC 3 – ROTARY PUMP SHORT QUESTIONS				
Sr. No.	**QUESTIONS**	**Marks**	**CO**	**RBT Level**
1	With neat sketch, explain in brief the working of Vane pump. **[NLJIET]**	3	CO4	UN
2	Explain the working principle of Vane pump with neat sketch? **[NLJIET]**	4	CO4	UN
3	With neat sketch, explain construction and working of Gear pump and Screw pump. **[NLJIET]**	4	CO4	UN
4	Write short note on Gear Pump. **[NLJIET]**	4	CO4	UN
5	Explain any one type of rotary pumps with figure. **(Jan-2024) [NLJIET]**	4	CO4	UN
DESCRIPTIVE QUESTIONS				
Sr. No.	**QUESTIONS**	**Marks**	**CO**	**RBT Level**
1	What do you mean by Positive displacement pump? Explain briefly any two rotary pumps with its application. **[NLJIET]**	5	CO4	UN

2	Classify the rotary pumps and describe with neat sketch, working of a Rotary gear pump. **[NLJIET]**	**6**	**CO4**	**UN**
3	With neat sketch, explain construction and working of Gear pump and Screw pump. **[NLJIET]**	**7**	**CO4**	**UN**

UNIT: 6

REFRIGERATION AND AIR CONDITIONING

Refrigerant, Vapor compression refrigeration system, Vapor absorption refrigeration system, Domestic Refrigerator, Window and split air conditioners.

TOPIC 1 – CONCEPTS OF REFRIGERATION

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Define the terms 'Refrigeration', 'Ton of Refrigeration' and 'Coefficient of Performance'. **(Jan-2020) [NLJIET]**	**3**	**CO4**	**RM**
2	Explain: 1 ton of refrigeration and refrigeration effect. **(Aug-2022) [NLJIET]**	**3**	**CO4**	**RM**
3	Discuss the operational difference between Vapour compression and Vapour absorption refrigeration cycle. **(Jun-2019) [NLJIET]**	**3**	**CO4**	**UN**
4	Discuss the term: (1) Condenser (2) Baffle tray (3) Evaporator **(Jan-2019) [NLJIET]**	**3**	**CO4**	**RM**
5	Draw neat sketch of domestic refrigerator and list its different parts. **(Mar-2021) [NLJIET]**	**3**	**CO4**	**RM**
6	Define the terms: Refrigeration, Refrigeration effect and Coefficient of Performance. **(Jan-2025-New) [NLJIET]**	**3**	**CO4**	**RM**
7	Define the terms 'Refrigeration', 'Ton of Refrigeration' and 'Coefficient of Performance'. **(Nov-2020) [NLJIET]**	**4**	**CO4**	**RM**
8	Define - One ton of refrigeration; COP **(Mar-2022) [NLJIET]**	**4**	**CO4**	**RM**
9	With neat sketch, describe the working of simple vapour compression refrigeration cycle. (Drawing p-h and T-S chart) **[NLJIET]**	**4**	**CO4**	**UN**

10	Draw a schematic diagram of vapour compression refrigeration system. State the function of each main component. **(Jul-2023) [NLJIET]**	**4**	**CO4**	**RM**
11	Give comparison between vapour compression and vapour absorption system. **[NLJIET]**	**4**	**CO4**	**UN**
12	Draw a schematic diagram of the vapor absorption refrigeration system. **(Jan-2024) [NLJIET]**	**4**	**CO4**	**RM**
13	Explain working of Vapor absorption refrigeration system with neat sketch. **(Jun-2025-New) [NLJIET]**	**4**	**CO4**	**UN**
14	What is refrigerant? Describe the properties of good refrigerant. **(Aug-2022) [NLJIET]**	**4**	**CO4**	**RM**
15	What is refrigerant? Describe the properties of good refrigerant. **(Jul-2025) [NLJIET]**	**4**	**CO4**	**RM**
16	List the various Refrigerant used in refrigeration system. Also write the desirable properties of refrigerant. **[NLJIET]**	**4**	**CO4**	**RM**
17	Write short note on Domestic refrigerator. **[NLJIET]**	**4**	**CO4**	**UN**
18	Explain with neat diagram, the working of Domestic refrigerator. **(Mar-2021) [NLJIET]**	**4**	**CO4**	**RM**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Draw working fluid flow diagram of the Vapour Compression Refrigeration System and describe the function of each important component of the system. **[NLJIET]**	**5**	**CO4**	**RM**
2	What should be properties of common refrigerants? **[NLJIET]**	**5**	**CO4**	**RM**
3	Explain with a neat sketch, the working of a vapour compression refrigerator. **[NLJIET]**	**5**	**CO4**	**UN**
4	Explain Vapour absorption Refrigeration system with the neat sketch. **[NLJIET]**	**6**	**CO4**	**UN**
5	Draw line diagram of vapour compression refrigeration cycle and represent on P-h and T-S diagram and state function of individual components of vapour compression refrigeration system. **[NLJIET]**	**7**	**CO4**	**RM**

6	Explain with neat sketch, vapor compression refrigeration cycle. What is C. O. P.? **[NLJIET]**	**7**	**CO4**	**UN**
7	Explain Vapor Compression Refrigeration system with neat sketch. Also draw p-h and T-s diagram for the same. **(Jan-2020) [NLJIET]**	**7**	**CO4**	**UN**
8	What is Refrigeration? Explain the working of vapour compression refrigeration cycle. Name basic components of VCRC. **[NLJIET]**	**7**	**CO4**	**UN**
9	Mention different parts of vapor compression refrigeration cycle along with their functions. Also draw a neat diagram of vapor compression refrigeration cycle. **[NLJIET]**	**7**	**CO4**	**UN**
10	Explain vapour compression refrigeration cycle used in domestic refrigerator. **(Mar-2022) [NLJIET]**	**7**	**CO4**	**UN**
11	Explain vapour compression refrigeration cycle used in domestic refrigerator. **(Jul-2025) [NLJIET]**	**7**	**CO4**	**UN**
12	Explain vapour compression refrigeration cycle. **(Mar-2023) [NLJIET]**	**7**	**CO4**	**UN**
13	Define the term 'Refrigeration'. Explain Vapour compression refrigerator. **[NLJIET]**	**7**	**CO4**	**UN**
14	Explain with neat sketch vapour compression refrigeration system. **(Jan-2025-New) [NLJIET]**	**7**	**CO4**	**UN**

NUMERICAL

Sr. No.	QUESTION	Marks	CO	RBT Level
1	It is required to produce ice from 36 deg water. The capacity of the ice plant is 6 ton and specific heat of the water is 4.18 kJ/kg.K. Determine the amount of ice produced in 3 hours. **(Jun-2019) [NLJIET]**	**4**	**CO4**	**AN**

TOPIC 2 – CONCEPTS OF AIR CONDITIONING

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	What is split air conditioner? State its advantages over window air conditioner. **[NLJIET]**	**3**	**CO4**	**RM**

2	Explain working of split air conditioner. **(Mar-2021) [NLJIET]**	**3**	**CO4**	**UN**
3	Define air conditioning. Also give the classification of air conditioning system. **(Jan-2025-New) [NLJIET]**	**3**	**CO4**	**RM**
4	Define Air conditioning. State the basic components of air conditioning systems. **[NLJIET]**	**4**	**CO4**	**RM**
5	Draw a self-explanatory diagram of Window room air-conditioner. **(Jan-2024) [NLJIET]**	**4**	**CO4**	**RM**
6	Explain window air conditioner along with its advantages. **[NLJIET]**	**4**	**CO4**	**UN**
7	Explain working of split air conditioner. **(Jun-2025-New) [NLJIET]**	**4**	**CO4**	**UN**
8	Draw a labelled diagram of split air-conditioner. Why does the split air conditioner consume more power than a window air-conditioner of same capacity? **(Jul-2023) [NLJIET]**	**4**	**CO4**	**UN**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Explain Window air conditioner with neat sketch. **[NLJIET]**	**5**	**CO4**	**UN**
2	Define Air conditioning. List the important components of Air conditioning system. Also classify the system briefly. **[NLJIET]**	**5**	**CO4**	**RM**
3	With neat sketch, explain the working of Window Air Conditioner. **[NLJIET]**	**7**	**CO4**	**UN**
4	Explain with neat sketch, split air conditioner. State its advantages. **[NLJIET]**	**7**	**CO4**	**UN**
5	What is split AC? How it works? Explain with advantage and disadvantage. **(Jan-2019) [NLJIET]**	**7**	**CO4**	**UN**
6	Explain with neat sketch, construction and working of split air conditioner. **(Jan-2025) [NLJIET]**	**7**	**CO4**	**UN**
7	With neat sketch, explain construction and working of Window air-conditioner. **[NLJIET]**	**8**	**CO4**	**UN**

NUMERICALS				
Sr. No.	**QUESTIONS**	**Marks**	**CO**	**RBT Level**
1	Calculate the energy consumed in one month for following conditions: COP of air-conditioning unit = 5. Capacity of air conditioner = 2 TR. No of air conditioners = 8. All air conditioners run for 4 hours/day. **(Jun-2019) [NLJIET]**	**4**	**CO4**	**AN**
2	Calculate the energy consumed in one month for following conditions: COP of air-conditioning unit = 5. Capacity of air conditioner = 2 TR. No of air conditioners = 8. All air conditioners run for 4 hours/day. **(Nov-2020) [NLJIET]**	**7**	**CO4**	**AN**

UNIT: 7

COUPLINGS, CLUTCHES AND BRAKES AND TRANSMISSION OF MOTION AND POWER

Sub Unit 7.1 Couplings, Clutches and Brakes

Construction and applications of Couplings (Box; Flange; Pin type flexible; Universal and Oldham), Clutches (Disc and Centrifugal), and Brakes (Block; Shoe; Band and Disc)

TOPIC 1 – CONCEPTS OF COUPLINGS

SHORT QUESTIONS

Sr. No.	**QUESTIONS**	**Marks**	**CO**	**RBT Level**
1	What is the difference between rigid coupling and flexible coupling? **(Aug-2022) [NLJIET]**	**3**	**CO5**	**UN**
2	What is the difference between rigid coupling and flexible coupling? **(Jun-2025-New) [NLJIET]**	**3**	**CO5**	**UN**
3	State the need of couplings along with the purposes they serve in machineries. **(Jul-2023) [NLJIET]**	**3**	**CO5**	**RM**
4	Draw a half-sectional view of the Sleeve (muff) coupling. **(Jul-2023) [NLJIET]**	**3**	**CO5**	**RM**
5	What is function of Coupling? Name only various types of couplings. Explain Oldham coupling. **[NLJIET]**	**4**	**CO5**	**UN**

<table>
<tr><td>6</td><td>Explain Oldham’s coupling with neat sketch. [NLJIET]</td><td>4</td><td>CO5</td><td>UN</td></tr>
<tr><td>7</td><td>Explain Universal coupling with neat sketch. [NLJIET]</td><td>4</td><td>CO5</td><td>UN</td></tr>
<tr><td>8</td><td>Match the following. (Jan-2024) [NLJIET]
<table>
<tr><td>Rigid flange coupling</td><td>Axis of shafts are parallel but not in alignment.</td></tr>
<tr><td>Pin type flexible coupling</td><td>Where angular misalignment is more.</td></tr>
<tr><td>Universal coupling</td><td>Requires proper alignment of shaft axes.</td></tr>
<tr><td>Oldham coupling</td><td>For small angular misalignment.</td></tr>
</table></td><td>4</td><td>CO5</td><td>UN</td></tr>
<tr><td>9</td><td>What is coupling? Classify it. (Jan-2025) [NLJIET]</td><td>4</td><td>CO5</td><td>UN</td></tr>
<tr><th colspan="5">DESCRIPTIVE QUESTIONS</th></tr>
<tr><th>Sr. No.</th><th>QUESTIONS</th><th>Marks</th><th>CO</th><th>RBT Level</th></tr>
<tr><td>1</td><td>With a neat sketch, explain any one type of coupling. [NLJIET]</td><td>7</td><td>CO5</td><td>UN</td></tr>
<tr><td>2</td><td>Explain Flange coupling with neat sketch. [NLJIET]</td><td>7</td><td>CO5</td><td>UN</td></tr>
<tr><td>3</td><td>Classify the types of couplings and with neat figure, discuss the working of any coupling of your choice. [NLJIET]</td><td>7</td><td>CO5</td><td>UN</td></tr>
<tr><th colspan="5">TOPIC 2 – CONCEPTS OF CLUTCHES
SHORT QUESTIONS</th></tr>
<tr><th>Sr. No.</th><th>QUESTIONS</th><th>Marks</th><th>CO</th><th>RBT Level</th></tr>
<tr><td>1</td><td>Distinguish between coupling and clutch. (Nov-2020) [NLJIET]</td><td>3</td><td>CO5</td><td>UN</td></tr>
<tr><td>2</td><td>With neat sketch, explain construction and working of Cone clutch. [NLJIET]</td><td>3</td><td>CO5</td><td>UN</td></tr>
<tr><td>3</td><td>Explain with sketch: Centrifugal clutch. (Jan-2019) [NLJIET]</td><td>3</td><td>CO5</td><td>UN</td></tr>
<tr><td>4</td><td>What is clutch? State its functions. (Aug-2022) [NLJIET]</td><td>3</td><td>CO5</td><td>RM</td></tr>
<tr><td>5</td><td>Classify clutches. (Jan-2024) [NLJIET]</td><td>3</td><td>CO5</td><td>RM</td></tr>
<tr><td>6</td><td>Explain - Centrifugal clutch. [NLJIET]</td><td>4</td><td>CO5</td><td>UN</td></tr>
<tr><td>7</td><td>With simple sketch, explain working of Disc clutch. [NLJIET]</td><td>4</td><td>CO5</td><td>UN</td></tr>
<tr><td>8</td><td>Write a short note on a single plate (disc) friction clutch. (Jul-2025) [NLJIET]</td><td>4</td><td>CO5</td><td>UN</td></tr>
</table>

DESCRIPTIVE QUESTIONS				
Sr. No.	**QUESTIONS**	**Marks**	**CO**	**RBT Level**
1	Explain with neat sketch, the working of Single plate friction clutch. **[NLJIET]**	**5**	**CO5**	**UN**
2	What are the different types of couplings? Explain the Centrifugal clutch. **[NLJIET]**	**7**	**CO5**	**UN**
3	Write a short note on a Single plate (disc) friction clutch. **(Mar-2022) [NLJIET]**	**7**	**CO5**	**UN**
4	Classify different types of Clutch. Describe Single plate clutch with neat sketch. **[NLJIET]**	**7**	**CO5**	**UN**
5	Using neat sketch, explain the working of Cone clutch and Centrifugal clutch. **(Jun-2019) [NLJIET]**	**7**	**CO5**	**UN**

TOPIC 3 – CONCEPTS OF BRAKES

SHORT QUESTIONS				
Sr. No.	**QUESTIONS**	**Marks**	**CO**	**RBT Level**
1	Draw a labeled diagram of a shoe brake used in automobiles. **(Jan-2024) [NLJIET]**	**3**	**CO5**	**UN**
2	Give the similarities and differences between a clutch and a brake. **(Jan-2024) [NLJIET]**	**3**	**CO5**	**UN**
3	State the function of brakes and classify them. **(Jul-2023) [NLJIET]**	**3**	**CO5**	**RM**
4	Write function of clutch, brake and coupling. **(Mar-2023) [NLJIET]**	**3**	**CO5**	**RM**
5	Write function of Clutch, Brake and Coupling. **(Jul-2025) [NLJIET]**	**3**	**CO5**	**RM**
6	Define brake and also state difference between clutch and coupling. **(Jun-2025-New) [NLJIET]**	**3**	**CO5**	**UN**
7	Give the classification of brake and describe with neat sketch, the working principle of an internal expanding shoe brake. **[NLJIET]**	**4**	**CO5**	**UN**

8	Explain with neat sketch, the working of Internal Expanding Shoe Brake. **(Jan-2019) [NLJIET]**	4	CO5	UN
9	What is coupling? Explain Internal expanding shoe brake with a neat sketch? **(Mar-2023) [NLJIET]**	4	CO5	UN
10	Differentiate between Clutch and Brake. **[NLJIET]**	4	CO5	UN
11	Explain in brief: Muff coupling; Band brake. **[NLJIET]**	4	CO5	UN
12	Draw neat and labelled sketches only of following: Protected flange coupling, Internal expanding shoe brake, Band brake. **[NLJIET]**	4	CO5	RM
13	Draw neat and labelled sketches only of following: Protected flange coupling, Internal expanding shoe brake, Single plate friction clutch and Band brake. **[NLJIET]**	4	CO5	RM
14	Differentiate between clutch and brake. **(Jan-2025) [NLJIET]**	4	CO5	UN
15	Explain with a neat sketch working of internal expanding shoe brake. **(Jan-2025-New) [NLJIET]**	4	CO5	UN

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Explain with suitable diagram, working principle of Disc clutch and Band brake. **[NLJIET]**	5	CO5	UN
2	Sketch Single block brake, Double block brake and Band brake. Give their practical application. **[NLJIET]**	5	CO5	RM
3	What is the function of a brake? Explain with neat sketch, the working of an Internal expanding shoe brake. **[NLJIET]**	6	CO5	UN
4	What is brake? Describe an Internal expanding shoe brake with a neat sketch and state its applications. **[NLJIET]**	7	CO5	UN
5	Explain about Brakes, Clutches and Couplings. **[NLJIET]**	7	CO5	UN
6	Explain: (i) Muff coupling (ii) Single plate clutch (iii) Band brake. **[NLJIET]**	7	CO5	UN

7	Sketch-Single block brake, Double block brake and Band brake. Give their practical application. **[NLJIET]**	**7**	**CO5**	**RM**
8	Using neat sketch, explain the working of Block brake and Internal expanding shoe brake. **(Jun-2019) [NLJIET]**	**7**	**CO5**	**UN**
9	Draw sketch and label the following: Shoe brake, Cone clutch, Flange coupling. **(Mar-2021) [NLJIET]**	**7**	**CO5**	**RM**
10	Classify: (1) Clutches (2) Brakes. **(Jul-2024) [NLJIET]**	**7**	**CO5**	**RM**
11	What do you mean by the word 'Friction clutches'? Explain with neat sketch working of a centrifugal clutch. **(Jan-2025-New) [NLJIET]**	**7**	**CO5**	**UN**

UNIT: 7

COUPLINGS, CLUTCHES AND BRAKES AND TRANSMISSION OF MOTION AND POWER

Sub Unit 7.2 Transmission of Motion and Power

Shaft and axle, Different arrangement and applications of Belt drive; Chain drive; Friction drive and Gear drive.

TOPIC 1 – BASICS OF MECHANICAL DRIVES

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Compare individual drive and group drive. **[NLJIET]**	**4**	**CO5**	**UN**
2	What are different elements to transfer motion and power? Explain any one with neat sketch. **[NLJIET]**	**4**	**CO5**	**UN**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Write in detail about transmission of motion&power. **[NLJIET]**	**7**	**CO5**	**UN**
2	Discuss various types of power transmission devices. **[NLJIET]**	**7**	**CO5**	**UN**

TOPIC 2 – CONCEPTS OF BELT DRIVES AND PULLEY DRIVES

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Draw neat figures of any three types of belt drive. **(Mar-2023) [NLJIET]**	**3**	**CO5**	**RM**

2	Draw a schematic diagram of the Fast and Loose pulley drive. **(Jan-2024) [NLJIET]**	**3**	**CO5**	**RM**
3	Enlist various belt drives. Name any three belt materials. **(Jul-2023) [NLJIET]**	**3**	**CO5**	**RM**
4	Enlist types of belt drive. **(Jan-2025) [NLJIET]**	**3**	**CO5**	**RM**
5	Describe briefly using neat diagrams, the types of belt drives. **(Jan-2020) [NLJIET]**	**4**	**CO5**	**RM**
6	Explain types of Belt Drives. **(Mar-2022) [NLJIET]**	**4**	**CO5**	**UN**
7	Explain with neat sketch, the working of Fast and Loose pulley. **(Mar-2023) [NLJIET]**	**4**	**CO5**	**UN**
8	Define Velocity ratio of pulleys and discuss effect of slip and creep on motion transmission. **(Nov-2020) [NLJIET]**	**4**	**CO5**	**RM**

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Enlist various types of pulleys used for power drives. Explain any two with neat sketch and their application. **[NLJIET]**	**5**	**CO5**	**UN**
2	What are the materials used for belts. Compare Flat and V – belt drive. **[NLJIET]**	**6**	**CO5**	**UN**
3	What are belt drives? List various belt drives and explain Cross belt drive. **[NLJIET]**	**6**	**CO5**	**UN**
4	Draw neat and labeled sketches of following – 1. Open belt drive 2. Quarter twist drive 3. Fast and loose pulley drive 4. Stepped pulley drive. **[NLJIET]**	**7**	**CO5**	**RM**
5	What are belt drives? List various belt drives and explain Cross belt drive. **[NLJIET]**	**7**	**CO5**	**UN**
6	Write short note on all types of belt drive. **[NLJIET]**	**7**	**CO5**	**UN**
7	Define the following terms related to belt drive: 1. Velocity ratio 2. Initial Tension 3. Slip 4. Creep 5. Power transmitted in belt drive **[NLJIET]**	**7**	**CO5**	**RM**

TOPIC 3 – CONCEPTS OF CHAIN DRIVES AND GEAR DRIVES

SHORT QUESTIONS

<table>
<tr><th>Sr. No.</th><th>QUESTIONS</th><th>Marks</th><th>CO</th><th>RBT Level</th></tr>
<tr><td>1</td><td>Differentiate between a belt drive and gear drive. (Jan-2024) [NLJIET]</td><td>3</td><td>CO5</td><td>UN</td></tr>
<tr><td>2</td><td>Explain in brief: (i) Worm gear. [NLJIET]</td><td>3</td><td>CO5</td><td>UN</td></tr>
<tr><td>3</td><td>Explain difference between belt drive and chain drive. (Mar-2023) [NLJIET]</td><td>3</td><td>CO5</td><td>UN</td></tr>
<tr><td>4</td><td>Match the following: (Jul-2023) [NLJIET]
<table>
<tr><td>Spiral gears</td><td>Parallel shafts</td></tr>
<tr><td>Bevel gears</td><td>Intersecting shafts</td></tr>
<tr><td>Worm gears</td><td>Non-parallel non-intersecting shafts</td></tr>
<tr><td>Helical gears</td><td>Shaft axes at right angles & non-intersecting</td></tr>
</table></td><td>3</td><td>CO5</td><td>UN</td></tr>
<tr><td>5</td><td>State the application of following gears 1. Helical gear, 2. Worm and worm gear 3. Rack and pinion. (Jan-2025) [NLJIET]</td><td>3</td><td>CO5</td><td>RM</td></tr>
<tr><td>6</td><td>Compare belt drive, chain drive and gear drive. (Mar-2022) [NLJIET]</td><td>4</td><td>CO5</td><td>UN</td></tr>
<tr><td>7</td><td>Compare belt and gear drive. [NLJIET]</td><td>4</td><td>CO5</td><td>UN</td></tr>
<tr><td>8</td><td>Write short note on Helical gear. [NLJIET]</td><td>4</td><td>CO5</td><td>UN</td></tr>
<tr><td>9</td><td>List advantages and disadvantages of gear drive. [NLJIET]</td><td>4</td><td>CO5</td><td>RM</td></tr>
<tr><td>10</td><td>What do you understand by gear train?
Discuss various types of gear train. [NLJIET]</td><td>4</td><td>CO5</td><td>UN</td></tr>
<tr><td>11</td><td>Sketch and describe helical and bevel gear and state applications of each. [NLJIET]</td><td>4</td><td>CO5</td><td>UN</td></tr>
<tr><td>12</td><td>Explain with neat sketch, Worm and worm wheel. [NLJIET]</td><td>4</td><td>CO5</td><td>UN</td></tr>
<tr><td>13</td><td>Define pitch circle, diametric pitch, addendum and dedendum using neat sketch. (Jun-2019) [NLJIET]</td><td>4</td><td>CO5</td><td>RM</td></tr>
<tr><td>14</td><td>Write down comparison between Belt, Chain and Gear drive. (Jan-2025-New) [NLJIET]</td><td>4</td><td>CO5</td><td>UN</td></tr>
</table>

DESCRIPTIVE QUESTIONS				
Sr. No.	**QUESTIONS**	**Marks**	**CO**	**RBT Level**
1	Discuss the relative merits and demerits of belt, chain and gear drives. **[NLJIET]**	**5**	**CO5**	**UN**
2	State the application, advantages and disadvantages of 1. Belt drive 2. Chain drive 3. Gear drive **[NLJIET]**	**6**	**CO5**	**RM**
3	Compare: Belt, Chain and Gear drives. **(Jul-2024) [NLJIET]**	**7**	**CO5**	**UN**
4	Compare belt drive, chain drive and gear drive. **(Jul-2025) [NLJIET]**	**7**	**CO5**	**UN**
5	Explain in brief - Worm gears; Rack and pinion; Crossed belt drive. **[NLJIET]**	**7**	**CO5**	**UN**
6	Explain with neat sketch, the working of belt drives and gear drives. **[NLJIET]**	**7**	**CO5**	**UN**
7	Classify Mechanical drives. Explain in brief: 1. Cross belt drive 2. Helical gear **[NLJIET]**	**7**	**CO5**	**UN**
8	Compare belt drive, chain drive and gear drive based on following criteria: 1. Main elements 2. Application suitability w.r.t to distance and velocity ratio 3. Space requirement 4. Slip 5. Design & Manufacturing complexity 6. Life 7. Maintenance. **[NLJIET]**	**7**	**CO5**	**UN**

UNIT: 8
ENGINEERING MATERIALS

Types, properties and applications of Ferrous & Nonferrous metals, Timber, Abrasive material, silica, ceramics, glass, graphite, diamond, plastic and polymer.

SHORT QUESTIONS				
Sr. No.	**QUESTIONS**	**Marks**	**CO**	**RBT Level**
1	Explain physical properties of engineering materials. **(Aug-2022) [NLJIET]**	**3**	**CO5**	**UN**

2	Write short note on**:** Composite materials. **[NLJIET]**	**3**	**CO5**	**UN**
3	List the important properties of engineering materials. **(Jul-2024) [NLJIET]**	**3**	**CO5**	**RM**
4	Define ferrous and nonferrous material with example. **(Mar-2022) [NLJIET]**	**3**	**CO5**	**RM**
5	Define the following mechanical properties. (i) Ductility (ii) Hardness (iii) Plasticity **(Mar-2023) [NLJIET]**	**3**	**CO5**	**RM**
6	Define elasticity, rigidity, hardness, fatigue, ductility, brittleness. **[NLJIET]**	**3**	**CO5**	**RM**
7	Define the following mechanical properties of metal - 1. Elasticity 2. Brittleness 3. Hardness **[NLJIET]**	**3**	**CO5**	**RM**
8	Define: (i) Creep (ii) Toughness (iii) Fatigue **[NLJIET]**	**3**	**CO5**	**RM**
9	Define following material properties. 1. Ductility 2. Plasticity 3. Malleability**. (Jun-2019) [NLJIET]**	**3**	**CO5**	**RM**
10	Classify properties of engineering material. **[NLJIET]**	**3**	**CO5**	**RM**
11	Define the following mechanical properties (i) Elasticity (ii) Toughness (iii) Ductility **[NLJIET]**	**3**	**CO5**	**RM**
12	Explain classification of engineering materials. **(Mar-2023) [NLJIET]**	**3**	**CO5**	**UN**
13	How metals are classified? Show with block diagram. **(Nov-2020) [NLJIET]**	**3**	**CO5**	**RM**
14	Enlist physical properties of Engineering materials. **[NLJIET]**	**4**	**CO5**	**RM**
15	Write about engineering materials. **[NLJIET]**	**4**	**CO5**	**UN**
16	Define Ductility, Elasticity, Plasticity and Weldability. **[NLJIET]**	**4**	**CO5**	**RM**
17	What do you understand by non-metallic materials? Name any six and state their practical importance. **[NLJIET]**	**4**	**CO5**	**RM**
18	List properties of copper and its applications. **(Mar-2021) [NLJIET]**	**4**	**CO5**	**RM**
19	What are nonferrous metals? Name any five and state their application. **(Aug-2022) [NLJIET]**	**4**	**CO5**	**RM**

20	Define: Malleability, Compressive strength, Toughness and Brittleness. **[NLJIET]**	**4**	**CO5**	**RM**
21	Define – Hardness, Creep, Resilience, Toughness. **[NLJIET]**	**4**	**CO5**	**RM**
22	Define ferrous and nonferrous materials with their properties and suitable application. **[NLJIET]**	**4**	**CO5**	**RM**
23	Define the terms: (i) Hardness (ii) Toughness (iii) Ductility (iv) Elasticity. **(Jan-2020) [NLJIET]**	**4**	**CO5**	**RM**
24	Define the terms: Hardness, Toughness, Ductility, Elasticity. **(Jun-2025-New) [NLJIET]**	**4**	**CO5**	**RM**
25	Discuss briefly any two non-metallic materials. **[NLJIET]**	**4**	**CO5**	**UN**
26	Write a short note on: Engineering Materials. **[NLJIET]**	**4**	**CO5**	**UN**
27	Identify the following properties of materials: 1) This property is desirable for parts subjected to impact loads and vibrations. 2) By this property gold, silver, can be flattened or bent without cracking when hammered. 3) This property is useful for materials subjected to high temperatures like boilers and turbines. 4) This property is necessary for a material to be used in making springs. **(Jul-2023) [NLJIET]**	**4**	**CO5**	**UN**
28	Provide meaning of the following statements regarding cast iron material: 1) It is brittle and has low ductility. 2) It has high compressive strength but low tensile strength. 3) It has excellent castability. 4) It has poor formability. **(Jan-2024) [NLJIET]**	**4**	**CO5**	**UN**
29	What do you mean by non-metallic materials? Name any three non-metallic materials and state its practical importance. **(Jan-2025-New) [NLJIET]**	**4**	**CO5**	**RM**

DESCRIPTIVE QUESTIONS

Sr. No.	**QUESTIONS**	**Marks**	**CO**	**RBT Level**
1	Discuss briefly alloy steels and give its practical application. **[NLJIET]**	**5**	**CO5**	**UN**
2	Describe in brief the various non-ferrous metals along with their applications. **[NLJIET]**	**6**	**CO5**	**UN**
3	State three engineering application of following materials, (i) Diamond (ii) Composite materials. **[NLJIET]**	**6**	**CO5**	**RM**
4	Describe in brief the various non-ferrous metals along with their applications. **[NLJIET]**	**7**	**CO5**	**UN**

5	Write note on the following engineering materials: (i) Mild steel (ii) Plywood (iii) Fireclay. **[NLJIET]**	**7**	**CO5**	**UN**
6	What is the difference between ferrous and nonferrous materials? List out various ferrous and nonferrous materials with their application. **[NLJIET]**	**7**	**CO5**	**UN**
7	Enlist the properties and applications of any one non-ferrous metal. **[NLJIET]**	**7**	**CO5**	**RM**
8	Define following mechanical properties – Elasticity, Malleability, Ductility, Impact strength, Hardness, Toughness, Resilience **[NLJIET]**	**7**	**CO5**	**RM**
9	Define following mechanical properties – Elasticity, Malleability, Ductility, Stiffness, Hardness, Toughness, Resilience (**Mar-2022) [NLJIET]**	**7**	**CO5**	**RM**
10	Classify properties of engineering material. Explain any three of them. **[NLJIET]**	**7**	**CO5**	**UN**
11	Discuss the following with application and properties: 1. Glass 2. Ceramic 3. Plastics **(Nov-2020) [NLJIET]**	**7**	**CO5**	**UN**
12	Write note on the following engineering materials (i) Cast iron (ii) Timber (iii) Diamond **[NLJIET]**	**7**	**CO5**	**UN**
13	List and explain physical, thermal, mechanical and electrical properties of metals. **(Mar-2021) [NLJIET]**	**7**	**CO5**	**UN**
14	Explain following properties of materials. Elasticity, Plasticity, Ductility, Brittleness, Hardness, Toughness, Fatigue. **(Jan-2025) [NLJIET]**	**7**	**CO5**	**UN**
15	Define following mechanical properties: (1) Elasticity (2) Malleability (3) Ductility (4) Stiffness (5) Hardness (6) Toughness (7) Resilience. **(Jul-2025) [NLJIET]**	**7**	**CO5**	**RM**

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